Vanishing theorems and incomplete deformations for vacuum data in $\mathcal{N} = 4$ SYM

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Abstract

We formulate the behavior of 1-point functions in the source-free, symmetric vacuum of an exact conformal theory, taking $\mathrm{AdS}_5 \times S^5$ and its boundary dual $\mathcal{N} = 4$ $SU(N)$ super Yang–Mills theory as the reference system. As candidates for detecting nontrivial 1-point vacuum data we adopt the operator family $\mathcal{C}_{\mathrm{ex}} = \{\mathcal{O}^a_\Delta(x),\, J^A_\mu(x),\, T_{\mu\nu}(x),\, \partial^\mu J^A_\mu(x)\}$, where $\mathcal{O}^a_\Delta$ are local scalar primaries with $\Delta > 0$, J^A_μ are conserved currents, and $T_{\mu\nu}$ is the energy-momentum tensor.

As the first result, we prove a unified vanishing theorem: in the Poincaré-invariant, conformally invariant, source-free vacuum of $\mathcal{N} = 4$ SYM, the 1-point functions of all operators in $\mathcal{C}_{\mathrm{ex}}$ vanish. The individual vanishing results—via boundary Ward identities and bulk holographic renormalization—are separately well known; the unified formulation for the operator family $\mathcal{C}_{\mathrm{ex}}$ provides the basis for the systematic analysis that follows. Second, under deformations $S = S_{\mathrm{CFT}} + \Delta S_{\mathrm{def}}$—relevant deformations, finite temperature, finite chemical potential, vacuum selection, time-dependent backgrounds, etc.—we systematize the conditions under which $\mathcal{C}_{\mathrm{ex}}$-relative nonzero 1-point functions are permitted, and identify the deformations for which constructive realizations are known. The $\Lambda > 0$ background is placed in this framework as a geometric implementation of such deformations.

Third, we examine observables outside $\mathcal{C}_{\mathrm{ex}}$—defects (Wilson loops, etc.), information-theoretic quantities (entanglement entropy, mutual information), and higher-point correlators—and classify observables into four layers. The vanishing theorem is specific to Layer I (local 1-point data); in the standard example, Layers II–IV carry nontrivial data even in the symmetric vacuum (scope-delimitation theorem). Fourth, we unify the four-layer classification with the deformation theory and show that representative deformations trigger systematic responses across all four layers (layer-resolved response theorem). Of the 20 cells formed by representative deformations and the four layers, 19 are supported by literature evidence, explicit computation, or the QFT-on-de Sitter framework.

The arguments throughout rely on established results in quantum field theory and the AdS/CFT correspondence; the individual ingredients are separately known. What is new are structural facts that emerge only from their unification: the vanishing is sharply confined to Layer I, the deformation response across all four layers is systematic rather than ad hoc, and the sole unconfirmed cell ($\Lambda > 0\times$ Layer II) is precisely locatable. All theorems are independent of the completeness of $\mathcal{C}_{\mathrm{ex}}$. What remains open is not the correctness of the theorems but the completeness of the adopted

operator family $\mathcal{C}_{\rm ex}$ and its four-layer extension—that is, the *scope* of the claims made here.

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I. INTRODUCTION

It is well known that 1-point functions of local operators are strongly constrained in the symmetric vacuum of a conformal field theory (CFT). In a Poincaré-invariant, conformally

invariant vacuum, the 1-point function of a scalar primary with scaling dimension $\Delta > 0$ vanishes by the scale Ward identity [13, 14]; that of a conserved current vanishes by Lorentz invariance [13, 14]; and that of the energy-momentum tensor on a flat background vanishes by Lorentz invariance and the trace Ward identity [5, 13, 14]. On the bulk side of AdS/CFT, holographic renormalization shows that the normalizable coefficient in a source-free pure-AdS background vanishes, giving a geometric expression of the same conclusion [9–12].

However, these vanishing results are usually stated separately for each operator type, and no study has systematically addressed the following questions:

(i) Can these individual vanishing results be unified into a single theorem for a defined operator family?

(ii) When deformations—relevant deformations, finite temperature, finite density, vacuum selection, $\Lambda > 0$ backgrounds, etc.—are introduced, how is the vanishing lifted, and can this be classified systematically?

(iii) What is the scope of the vanishing theorem—does it hold for observables beyond local 1-point functions?

(iv) How do observables outside the scope of the vanishing theorem respond to deformations?

This paper provides systematic answers to these questions in the best-controlled standard example of AdS/CFT, namely $\mathrm{AdS}_5 \times S^5$ and its boundary dual $\mathcal{N} = 4$ $SU(N)$ super Yang–Mills theory (SYM) [1–4].

The main results of this paper are as follows.

Vanishing theorem. The 1-point functions of all operators in the operator family $\mathcal{C}_{\mathrm{ex}}$, operationally defined in §II, vanish in the source-free, symmetric vacuum of the exact conformal theory (Theorem III.18). The proof is given from both the boundary Ward identities [5, 6, 13, 14] and the bulk holographic renormalization [9–12].

Nonzero realization. Under a deformation ΔS_{def},

$$S = S_{\mathrm{CFT}} + \Delta S_{\mathrm{def}} \,, \tag{1}$$

—relevant deformations, finite temperature, finite chemical potential, vacuum selection, time-dependent backgrounds, etc.—at least one assumption supporting the vanishing theorem is violated, and $\mathcal{C}_{\text{ex}}$-relative nonzero 1-point functions are permitted (Theorem IV.16). The concrete realization depends on the type of deformation and the quantum numbers of the operator.

$\Lambda > 0$ **background.** The $\Lambda > 0$ background does not preserve the assumptions underlying the vanishing theorem, through the introduction of an IR scale by the Hubble parameter H, the Gibbons–Hawking temperature $T_{\text{dS}} = H/(2\pi)$ for a local observer, and the modification of the source/response dictionary [15–21]. Thus $\Lambda > 0$ is placed as a *geometric deformation* that breaks exact conformal invariance in a manner distinct from matter-sector Lagrangian deformations (Proposition V.14). The above results are collected in the unification theorem (Theorem VI.1).

Four-layer classification and scope delimitation. We examine observables outside $\mathcal{C}_{\text{ex}}$—Wilson loops and other defects, entanglement entropy, mutual information, and other information-theoretic quantities, and higher-point correlators—and classify observables into four layers (Definition VII.13): Layer I (local 1-point data), Layer II (defect / nonlocal data), Layer III (information-theoretic data), Layer IV (correlation data). The vanishing theorem is specific to Layer I; in the standard example $\text{AdS}_5 \times S^5/\mathcal{N} = 4$ SYM, Layers II–IV carry nontrivial data even in the symmetric vacuum (Theorem VII.16, scope-delimitation theorem).

Layer-resolved response. Unifying the four-layer classification with the deformation theory, we show that representative deformations—relevant deformation, finite temperature, finite density, Coulomb branch, and $\Lambda > 0$—trigger systematic responses across all four layers (Theorem VIII.11, layer-resolved response theorem). Of the 20 cells formed by representative deformations and the four layers, 19 are supported by literature evidence, explicit computation, or the QFT-on-de Sitter framework. The sole unconfirmed cell is $\Lambda > 0\times$ Layer II (defect), which originates from the framework dependence of the observable definition.

Scope caveat. All theorems of this work are independent of the completeness of $\mathcal{C}_{\text{ex}}$. What remains open is not the correctness of the theorems but the completeness of the adopted operator family $\mathcal{C}_{\text{ex}}$ and its four-layer extension—that is, the *scope* of the claims made here.

The arguments in this paper are built on standard results in quantum field theory and the AdS/CFT correspondence. The vanishing theorem (§III) relies on the CFT Ward identities [5, 6, 13, 14] and holographic renormalization [9–12], each of which is individually textbook material. The description of nonzero realization (§IV) uses finite-temperature and finite-density holographic computations [4, 39, 42], conformal perturbation theory [14], and the structure of the Coulomb branch [4, 6, 7]. The $\Lambda > 0$ discussion (§V) is based on the causal structure of de Sitter space and the Gibbons–Hawking temperature [15, 17, 20, 21]. The four-layer classification (§VII) employs the exact Wilson-loop result from Pestun localization [29–31], holographic entanglement entropy [32, 33], and results on the modular Hamiltonian [34]. The layer-resolved response (§VIII) combines the finite-temperature Wilson–Polyakov loop [40], extensions of holographic entanglement entropy to finite temperature and density [42], holographic entanglement entropy on the Coulomb branch [41], and changes in holographic entanglement entropy under RG flow [43].

The individual ingredients are separately known; the novelty lies in structural facts that emerge only from their integration. First, once the vanishing theorem is formulated for the operator family $\mathcal{C}_{\text{ex}}$ as a whole (Theorem III.18, Theorem VI.1), it admits a sharp scope boundary: vanishing is confined to Layer I, while Layers II–IV carry nontrivial data even in the exact conformal vacuum (Theorem VII.16). This dichotomy is not visible from any single vanishing result taken in isolation. Second, the "deformation $\times$ layer" matrix (Theorem VIII.11) makes visible that the deformation response across all four layers is systematic rather than ad hoc, with 19 of 20 cells confirmed. Third, the sole unconfirmed cell ($\Lambda > 0\times$ Layer II) is identifiable precisely because the matrix structure exists; without it, this gap would not be locatable.

The paper is organized as follows. In §II, we operationally define the exact conformal theory and the operator family $\mathcal{C}_{\text{ex}}$ of candidate 1-point observables, and delineate the propositions to be proved from the completeness question to be left open. In §III, we show that all 1-point functions of $\mathcal{C}_{\text{ex}}$ vanish in the conformal vacuum of $\mathcal{N} = 4$ SYM. In §IV, we systematize the conditions under which 1-point functions of $\mathcal{C}_{\text{ex}}$ can become nonzero under deformations ΔS_{def}. In §V, we place the $\Lambda > 0$ background as a geometric deformation. In §VI, we collect the results of §III–§V into the unification theorem. In §VII, we examine observables outside $\mathcal{C}_{\text{ex}}$, introduce the four-layer classification, and delimit the scope of the

vanishing theorem. In §VIII, we unify the four-layer classification with the deformation theory and establish the layer-resolved response theorem. In §IX, we condense the conclusions of this work into a final claim and state the falsification structure and the limits of scope.

We begin in §II by introducing the definitions that serve as the starting point.

II. DEFINITIONS AND FORMULATION

A. Purpose

The purpose of this paper is to formulate, in the most controlled standard example of AdS/CFT—namely $\mathrm{AdS}_5 \times S^5$ and its boundary dual $\mathcal{N} = 4$ $SU(N)$ SYM [1–4]—the proposition that **the quantities adopted in this work as candidate 1-point observables vanish as 1-point functions in the symmetric vacuum of an exact conformal theory, and nontrivial 1-point vacuum data emerge only through symmetry breaking, scale generation, or vacuum selection**.

What we call *nontrivial vacuum data* in this paper does not merely mean that operators can be defined on the Hilbert space. Nor does the nonvanishing of vacuum fluctuations or higher-point correlators automatically qualify as nontrivial vacuum data. What is at issue is **whether the vacuum itself carries a nontrivial bias, order, density, or production rate**, and this is determined fundamentally by the 1-point function. In this sense, on the CFT side the Ward identities and vacuum symmetry, and on the AdS side the structure of holographic 1-point functions, allow one to track rigorously what "vanishes in the symmetric vacuum" [2, 3, 5, 9, 10].

The central question of this paper can therefore be condensed to:

> *Can the symmetric vacuum of an exact conformal theory carry a nonzero 1-point function?*

Our answer is negative, relative to the adopted operator family. More precisely, in the conformal vacuum of $\mathcal{N} = 4$ SYM dual to $\mathrm{AdS}_5 \times S^5$, the 1-point functions of the operator family adopted as candidate 1-point observables vanish by symmetry and the Ward identities [1–7, 13, 14]. Nontrivial 1-point vacuum data therefore arise not from the exact conformal theory itself, but only through deformations, symmetry breaking, or non-symmetric vacuum selection.

B. The reference system $\mathrm{AdS}_5 \times S^5/\mathcal{N}=4$ SYM

To treat this problem with maximal rigor, we adopt type IIB string theory on $\mathrm{AdS}_5 \times S^5$ and its boundary dual, four-dimensional $\mathcal{N}=4$ SYM, as the reference system [1–4]. The reason is that this system implements the concept of an exact conformal theory most explicitly.

The boundary theory $\mathcal{N}=4$ SYM is exactly superconformal including quantum corrections, with

$$\beta(g_{\mathrm{YM}}) = 0\,, \tag{2}$$

so that the coupling constant does not run and the theory does not dynamically generate an internal scale [4, 6, 7]. The theory possesses maximal supersymmetry and is consistently defined under a large symmetry group including the R-symmetry [4, 6, 7]. The vacuum on a flat background is Poincaré-invariant and conformally invariant, providing a unique reference vacuum at zero source [4, 6, 13, 14].

On the bulk side, the corresponding vacuum is the maximally supersymmetric background $\mathrm{AdS}_5 \times S^5$; unless a source is turned on, the near-boundary data corresponding to 1-point functions of boundary operators are not excited [2, 3, 9–12]. Consequently, a **vanishing theorem based solely on symmetry** and a **geometric vanishing via holography** coincide.

This system is therefore the optimal testing ground for describing, without ambiguity, the relationship between exact conformal invariance and the vanishing of 1-point functions.

C. Definition of an exact conformal theory

In this paper, a theory satisfying all of the following conditions is called an **exact conformal theory**:

1. The RG flow of the coupling constant vanishes: $\beta(g) = 0$.

2. The vacuum is Poincaré-invariant.

3. The vacuum preserves scale or conformal symmetry.

4. No external scale has been introduced by an explicit relevant deformation.

5. In the source-free vacuum, no global symmetry is spontaneously broken.

Under this definition, the conformal vacuum of $\mathcal{N} = 4$ SYM on $\mathbb{R}^{1,3}$ is the prototypical example of an exact conformal theory [4, 6, 7, 13, 14]. The corresponding bulk description is the maximally symmetric vacuum of source-free type IIB string theory on $\mathrm{AdS}_5 \times S^5$ [1–4].

It should be emphasized that this definition of an exact conformal theory is not a subjective description of solvability or mathematical elegance, but an **operational concept specified by the Ward identities, vacuum symmetry, and the absence of scale generation** [6, 13, 14]. As we shall show, the vanishing of the 1-point functions treated in this paper follows mechanically from this definition.

D. Operational definition of 1-point vacuum data

In this paper, the presence or absence of nontrivial vacuum data is operationally determined by **the nonvanishing of the 1-point function** [2, 3, 5, 9, 10].

Concretely, 1-point vacuum data refers to any of the following:

- A nonzero scalar order parameter.
- A nonzero conserved charge density.
- A nonzero 1-point function of the energy-momentum tensor.
- A nonzero expectation value of the divergence of a current.

Phenomenologically, these correspond respectively to, e.g., mass generation, or the selection of an ordered phase; particle–antiparticle asymmetry or finite density; deviation from a homogeneous vacuum; and a production rate due to non-conservation processes.

However, when treating these within the framework of $\mathrm{AdS}_5 \times S^5/\mathcal{N} = 4$ SYM, we do not directly embed Standard Model operators. Since $\mathcal{N} = 4$ SYM contains neither fundamental quarks nor a Higgs doublet, it would be theoretically inaccurate to take, e.g., a baryon number operator or a Higgs VEV as direct targets [4, 7, 8].

Instead, we map 1-point vacuum data to an operator family that is rigorously definable within $\mathcal{N} = 4$ SYM:

$$\mathcal{O}^a_\Delta(x), \qquad J^A_\mu(x), \qquad T_{\mu\nu}(x), \qquad \partial^\mu J^A_\mu(x)\,, \tag{3}$$

and determine the presence or absence of nontrivial vacuum data by the vanishing or nonvanishing of their 1-point functions [5, 9–14].

This definition has two advantages. First, all quantities are rigorously defined as operators in the boundary CFT. Second, via the AdS/CFT dictionary, they can be geometrically re-expressed as near-boundary data of bulk fields [2, 3, 9–12].

E. Precise form of the claims

Let us now restate the claims of this paper precisely. What this paper demonstrates is not the vague assertion that "nothing happens in an exact conformal theory," nor the overly strong claims that "vacuum fluctuations do not exist" or "correlators vanish."

The claim of this paper takes the following specific form:

> *In the symmetric vacuum of an exact conformal theory, the 1-point functions of candidate 1-point observables belonging to the adopted operator family vanish.*

Thus, for example,

$$\langle T_{00}(x)\rangle = 0 \tag{4}$$

is asserted, but

$$\langle T_{00}(x)\, T_{00}(y)\rangle = 0 \tag{5}$$

is *not* asserted. Indeed, the latter is generically nonzero [5, 13, 14]. Likewise, the existence of vacuum fluctuations and higher-point correlators is distinguished from the nonvanishing of 1-point vacuum data.

This distinction is essential for this paper. The source-free, symmetric vacuum on $\mathrm{AdS}_5 \times S^5$ is quantum-mechanically nontrivial: it possesses a rich spectrum and nontrivial correlation functions, yet the vacuum itself carries no nonzero 1-point function [1–7, 9–14]. What is at issue in this paper is **not whether the quantum theory has content**, but **whether the vacuum carries nontrivial 1-point data**.

Furthermore, it is important that the correctness of the theorems in this paper does not depend on the adopted operator family "exhausting all necessary operators." What is rigorously shown is that **the 1-point function vanishes for each element of the chosen operator family**; the completeness of that operator family is left as a separate

open question. Thus, what is unresolved is not the correctness of the theorems, but their *scope*.

F. Deformations and nonzero 1-point functions

Given an exact conformal theory, we consider a deformation of the form

$$S = S_{\mathrm{CFT}} + \Delta S_{\mathrm{def}} \,. \tag{6}$$

Here ΔS_{def} is a deformation that violates at least one of the defining properties of the exact conformal theory; examples include

- relevant deformation,
- introduction of temperature or chemical potential,
- change of boundary conditions,
- vacuum selection,
- supersymmetry breaking,
- introduction of a curvature scale,
- time-dependent backgrounds,

among others [4, 7, 9–12, 17, 19–27].

In such cases, the vanishing of the 1-point functions stated in the previous section generically no longer holds. For instance, scalar sources or the excitation of normalizable modes permit $\langle \mathcal{O}_\Delta \rangle \neq 0$; finite-temperature or finite-density states permit $\langle T_{\mu\nu} \rangle \neq 0$ and $\langle J_\mu \rangle \neq 0$ [9–12]. Similarly, if a symmetry is broken by an anomaly or explicitly, $\langle \partial^\mu J_\mu \rangle$ can also become nonzero.

The structure of this paper is therefore twofold:

1. Show that in the symmetric vacuum of the exact conformal theory S_{CFT}, the 1-point functions of the candidate observables in the adopted operator family vanish.
2. Characterize how this vanishing is lifted by the deformation ΔS_{def}.

The first task is the primary concern of this paper, and it can be rigorously formulated using standard techniques of AdS/CFT [2, 3, 9, 10].

With these definitions in hand, we proceed in Section III to show that every 1-point function of $\mathcal{C}_{\rm ex}$ vanishes in the conformal vacuum.

III. RIGOROUS FORMULATION ON $\mathrm{AdS}_5 \times S^5$

In this section, we rigorously formulate the 1-point vacuum data introduced in Section II on the conformal vacuum of $\mathcal{N} = 4$ $SU(N)$ SYM and its $\mathrm{AdS}_5 \times S^5$ dual [2, 3, 5, 9–12]. The main objective is to show that, in the source-free symmetric vacuum of the exact conformal theory $S_{\rm CFT}$, the 1-point function of each element of the adopted operator family vanishes, from both (i) the boundary Ward identities and (ii) bulk holographic renormalization.

Throughout this section, the boundary spacetime is $\mathbb{R}^{1,3}$, the vacuum $|0\rangle$ is Poincaré-invariant and conformally invariant, and on the bulk side we adopt the source-free maximally supersymmetric solution $\mathrm{AdS}_5 \times S^5$ [1–4].

We emphasize that the correctness of the propositions proved in this section does not depend on the adopted operator family "exhausting all necessary operators." What is shown is that **the 1-point function vanishes for each element of the chosen operator family**; the completeness of that family is left as a separate open question.

A. Definition: the operator family of candidate 1-point observables

Following the discussion of Section II, we define the candidate 1-point observables to be treated rigorously in this paper as an operator family that is well-defined within $\mathcal{N} = 4$ SYM.

Definition III.1. *The operator family of candidate 1-point observables $\mathcal{C}_{\rm ex}$ (all local operators on the flat boundary $\mathbb{R}^{1,3}$) is defined by*

$$\mathcal{C}_{\rm ex} = \left\{ \mathcal{O}^a_\Delta(x),\ J^A_\mu(x),\ T_{\mu\nu}(x),\ \partial^\mu J^A_\mu(x) \right\}, \tag{7}$$

where

1. *$\mathcal{O}^a_\Delta(x)$ is a local scalar primary operator with scaling dimension $\Delta > 0$,*

2. *$J^A_\mu(x)$ is a conserved current corresponding to an exact global symmetry,*

3. *$T_{\mu\nu}(x)$ is the symmetric, conserved stress-energy tensor,*

4. *$\partial^\mu J^A_\mu(x)$ is the divergence of the current.*

The index a labels a basis of scalar operators with the same Δ, and A is a Lie algebra index of the global symmetry.

Remark III.2. The condition $\Delta > 0$ excludes the identity operator, whose expectation value is trivially unity and carries no vacuum-structure information.

Remark III.3. This definition does not directly embed Standard Model operators such as a baryon number operator or a Higgs doublet into $\mathcal{N} = 4$ SYM. What is treated here is the operator family that uniformly describes the abstract classes corresponding to their phenomenological roles—nonzero scalar 1-point function, nonzero charge density, nonzero stress tensor 1-point function, and nonzero current non-conservation [4, 7, 8].

Remark III.4. In this paper, $\mathcal{C}_{\text{ex}}$ is introduced as the operator family providing candidate 1-point observables. The theorems proved below are therefore $\mathcal{C}_{\text{ex}}$-relative statements. In particular, this paper shows

$$\forall\, \hat{\mathcal{O}} \in \mathcal{C}_{\text{ex}}, \qquad \langle 0|\hat{\mathcal{O}}|0\rangle = 0\,, \tag{8}$$

but does *not* claim that "all operators necessary for nontrivial vacuum data are exhausted by $\mathcal{C}_{\text{ex}}$." What is unresolved is the completeness of the operator family, which concerns not the correctness of the theorems but their scope.

B. Vanishing of 1-point functions: CFT side

We first establish the vanishing of 1-point functions on the boundary CFT side. The following propositions hold generally in any four-dimensional CFT with a Poincaré-invariant, conformally invariant vacuum, not only in $\mathcal{N} = 4$ SYM [5, 6, 13, 14].

Proposition III.5. *Let $\mathcal{O}_\Delta(x)$ be a local scalar primary with $\Delta > 0$. In a Poincaré-invariant, scale-invariant vacuum $|0\rangle$,*

$$\langle 0|\mathcal{O}_\Delta(x)|0\rangle = 0\,. \tag{9}$$

Proof. By translation invariance, $\langle 0|\mathcal{O}_\Delta(x)|0\rangle = C$ is a constant independent of x. Under a scale transformation $x \mapsto \lambda x$, a scalar primary transforms as $\mathcal{O}_\Delta(x) \mapsto \lambda^{-\Delta}\mathcal{O}_\Delta(\lambda x)$ [13, 14]. If the vacuum is scale-invariant, then

$$C = \lambda^{-\Delta}\, C \qquad (\forall\, \lambda > 0)\,.$$

For $\Delta > 0$, this forces $C = 0$. □

Corollary III.6. *In the conformal vacuum of $\mathcal{N} = 4$ SYM, for any nontrivial scalar primary $\mathcal{O}^a_\Delta$,*

$$\langle \mathcal{O}^a_\Delta(x)\rangle_0 = 0\,.$$

In particular, for the $\frac{1}{2}$-BPS scalar chiral primaries $\mathcal{O}_k^{(I_1\cdots I_k)} = \mathrm{Tr}\big(\Phi^{(I_1}\cdots\Phi^{I_k)}\big)$ – traces with $\Delta = k \geq 2$,

$$\langle \mathcal{O}_k^{(I_1\cdots I_k)}(x)\rangle_0 = 0\,.$$

See [4–7].

Proposition III.7. *Let $J^A_\mu(x)$ be a global symmetry current. In a Lorentz-invariant vacuum,*

$$\langle 0|J^A_\mu(x)|0\rangle = 0\,. \tag{10}$$

Proof. By translation invariance, $\langle J^A_\mu(x)\rangle_0 = v^A_\mu$ is a constant vector. In a Lorentz-invariant vacuum, the existence of a nonzero constant vector v^A_μ would single out a spacetime direction, contradicting Lorentz invariance [13, 14]. Hence $v^A_\mu = 0$. □

Corollary III.8. *In particular, the charge density vanishes: $\langle J^A_0(x)\rangle_0 = 0$.*

Proposition III.9. *In the conformal vacuum on a flat background $\mathbb{R}^{1,3}$,*

$$\langle 0|T_{\mu\nu}(x)|0\rangle = 0\,. \tag{11}$$

Proof. By translation and Lorentz invariance, $\langle T_{\mu\nu}(x)\rangle_0 = c\,\eta_{\mu\nu}$ for some constant c [13, 14]. In a four-dimensional CFT on a flat background, the trace Ward identity gives $\langle T^\mu{}_\mu(x)\rangle_0 = 0$ [5, 13, 14]. (In $\mathcal{N} = 4$ SYM, $a = c$; on flat $\mathbb{R}^{1,3}$ both the Euler density and the Weyl tensor vanish, so the trace anomaly is identically zero.) Therefore $0 = \eta^{\mu\nu}\langle T_{\mu\nu}\rangle_0 = 4c$, so $c = 0$ and $\langle T_{\mu\nu}(x)\rangle_0 = 0$. □

Corollary III.10. *In particular, the local energy density vanishes: $\langle T_{00}(x)\rangle_0 = 0$.*

Proposition III.11. *If $J^A_\mu(x)$ is a conserved current corresponding to an exact global symmetry, then the operator identity $\partial^\mu J^A_\mu(x) = 0$ holds, and therefore*

$$\langle \partial^\mu J^A_\mu(x) \rangle_0 = 0\,. \tag{12}$$

Proof. The Noether current of an exact symmetry is conserved. The vacuum of $\mathcal{N} = 4$ SYM does not break this conservation law; hence the result follows by taking the vacuum expectation value [4, 6, 7]. □

Remark III.12. The above result concerns exactly conserved currents. For anomalous or explicitly broken currents, the right-hand side may include an anomaly functional or source-dependent term. However, in the source-free conformal vacuum of $\mathcal{N} = 4$ SYM, the current sector of $\mathcal{C}_{\rm ex}$ is conserved and its 1-point function vanishes [4, 6, 7].

C. Generating-functional representation

The above propositions can also be expressed as variational statements of the generating functional. Introducing the boundary metric $g^{(0)}_{\mu\nu}$, scalar sources J_a, and vector sources A^A_μ, define the connected generating functional [2, 3, 5]

$$W[J_a, A^A_\mu, g^{(0)}_{\mu\nu}] = \log Z[J_a, A^A_\mu, g^{(0)}_{\mu\nu}]\,. \tag{13}$$

The 1-point functions are then

$$\langle \mathcal{O}^a_\Delta(x) \rangle = \frac{1}{\sqrt{-g^{(0)}(x)}} \frac{\delta W}{\delta J_a(x)}\,, \tag{14}$$

$$\langle J^\mu_A(x) \rangle = \frac{1}{\sqrt{-g^{(0)}(x)}} \frac{\delta W}{\delta A^A_\mu(x)}\,, \tag{15}$$

$$\langle T^{\mu\nu}(x) \rangle = \frac{2}{\sqrt{-g^{(0)}(x)}} \frac{\delta W}{\delta g^{(0)}_{\mu\nu}(x)}\,. \tag{16}$$

At zero source on a flat background, $J_a = 0$, $A^A_\mu = 0$, $g^{(0)}_{\mu\nu} = \eta_{\mu\nu}$, the results of §III B are summarized as

$$\left.\frac{\delta W}{\delta J_a(x)}\right|_0 = 0\,, \qquad \left.\frac{\delta W}{\delta A^A_\mu(x)}\right|_0 = 0\,, \qquad \left.\frac{\delta W}{\delta g^{(0)}_{\mu\nu}(x)}\right|_0 = 0\,, \tag{17}$$

where $|_0$ denotes evaluation at the above source-free, flat background.

D. Bulk side: holographic renormalization

We now rewrite the same propositions as statements about near-boundary data on the AdS side. By the AdS/CFT dictionary, the boundary generating functional equals the bulk renormalized on-shell action [2, 3, 9–11]:

$$W[J] = -S_{\text{ren}}^{\text{on-shell}}[\phi_{(0)} = J] \tag{18}$$

(in Euclidean signature). The 1-point functions of boundary operators are therefore given by variations of the renormalized on-shell action with respect to the boundary sources [9–12].

Proposition III.13. *Let the bulk scalar ϕ be dual to a boundary scalar operator $\mathcal{O}_\Delta$ of dimension Δ. Then*

$$\langle \mathcal{O}_\Delta(x) \rangle = -\frac{1}{\sqrt{g_{(0)}(x)}} \frac{\delta S_{\text{ren}}^{\text{on-shell}}}{\delta \phi_{(0)}(x)} \,. \tag{19}$$

In Fefferman–Graham coordinates,

$$ds^2 = \frac{L^2}{z^2}\big(dz^2 + g_{\mu\nu}(z,x)\, dx^\mu dx^\nu\big)\,, \qquad z \to 0\,, \tag{20}$$

the near-boundary expansion of ϕ reads

$$\phi(z,x) = z^{4-\Delta}\, \phi_{(0)}(x) + z^{\Delta}\, \phi_{(2\Delta-4)}(x) + \cdots\,, \tag{21}$$

and the renormalized 1-point function is

$$\langle \mathcal{O}_\Delta(x) \rangle = (2\Delta - 4)\, \phi_{(2\Delta-4)}(x) + \textit{local counterterm contributions} \tag{22}$$

[9–11].

Remark III.14. The numerical coefficient in (22) is convention-dependent. What is essential for this paper is the structure that the 1-point function is determined by the normalizable coefficient $\phi_{(2\Delta-4)}$ [9, 10].

Proposition III.15. *In the source-free pure-*AdS_5 *background, the scalar 1-point function vanishes:*

$$\phi_{(0)}(x) = 0 \quad \textit{and} \quad \phi_{(2\Delta-4)}(x) = 0 \quad \Longrightarrow \quad \langle \mathcal{O}_\Delta(x) \rangle = 0\,. \tag{23}$$

Proof. Under the source-free condition, the maximally symmetric pure-AdS_5 solution carries no bulk scalar hair. Consequently, the normalizable coefficient in the near-boundary expansion also vanishes [9–11]. The vanishing of the 1-point function then follows from the definition (22). □

Proposition III.16. *Let the bulk gauge field A^A_M be dual to the boundary current J^A_μ. Writing the near-boundary expansion as*

$$A^A_\mu(z,x) = A^{A(0)}_\mu(x) + z^2\, A^{A(2)}_\mu(x) + \cdots\,, \tag{24}$$

the renormalized current 1-point function is

$$\langle J^\mu_A(x)\rangle = -\frac{1}{\sqrt{g_{(0)}(x)}}\,\frac{\delta S^{\text{on-shell}}_{\text{ren}}}{\delta A^{A(0)}_\mu(x)} \propto A^{\mu(2)}_A(x) + \textit{local terms} \tag{25}$$

[9–11]. In the source-free pure-AdS_5 *background, $A^{A(0)}_\mu(x) = 0$ and $A^{A(2)}_\mu(x) = 0$, so $\langle J^\mu_A(x)\rangle = 0$.*

Proposition III.17. *For the bulk metric in the Fefferman–Graham expansion*

$$g_{\mu\nu}(z,x) = g_{(0)\mu\nu}(x) + z^2\, g_{(2)\mu\nu}(x) + z^4\, g_{(4)\mu\nu}(x) + \cdots\,, \tag{26}$$

the renormalized boundary stress tensor is [9–12]

$$\langle T_{\mu\nu}(x)\rangle = \frac{2}{\sqrt{-g_{(0)}(x)}}\,\frac{\delta W}{\delta g^{\mu\nu}_{(0)}(x)} = -\frac{2}{\sqrt{-g_{(0)}(x)}}\,\frac{\delta S^{\text{on-shell}}_{\text{ren}}}{\delta g^{\mu\nu}_{(0)}(x)}\,. \tag{27}$$

For a four-dimensional boundary, this takes the form

$$\langle T_{\mu\nu}\rangle = \frac{L^3}{4\pi G_5}\left(g_{(4)\mu\nu} + X_{\mu\nu}[g_{(0)}]\right), \tag{28}$$

where $X_{\mu\nu}[g_{(0)}]$ is determined by local curvature invariants of the boundary metric [9–12]. On a flat boundary $g_{(0)\mu\nu} = \eta_{\mu\nu}$, $X_{\mu\nu} = 0$, and for pure AdS_5*, $g_{(2)\mu\nu} = 0$, $g_{(4)\mu\nu} = 0$, so $\langle T_{\mu\nu}(x)\rangle = 0$ [9–12].*

E. Agreement between the CFT and bulk sides

Collecting the results above, the vanishing of 1-point functions from the boundary Ward identities and the vanishing of normalizable coefficients in the source-free pure-AdS_5 solution are two expressions of the same proposition [2, 3, 9–12].

Schematically, the correspondence reads

$$\langle \mathcal{O}_\Delta\rangle \;\longleftrightarrow\; \phi_{(2\Delta-4)}\,, \qquad \langle J^A_\mu\rangle \;\longleftrightarrow\; A^{A(2)}_\mu\,, \qquad \langle T_{\mu\nu}\rangle \;\longleftrightarrow\; g_{(4)\mu\nu}\,. \tag{29}$$

Thus, the vanishing of 1-point functions for the adopted operator family $\mathcal{C}_{\text{ex}}$ is expressed as the Ward identities on the boundary and as the absence of hair on the bulk side.

F. Main theorem

Combining the above, the main result of this section is as follows.

Theorem III.18 (Vanishing theorem)**.** *Consider the Poincaré-invariant, conformally invariant vacuum* $|0\rangle$ *of* $\mathcal{N}=4$ *SU(N) SYM on* $\mathbb{R}^{1,3}$*, and its AdS/CFT dual, the maximally supersymmetric vacuum of source-free type IIB theory on* $\mathrm{AdS}_5 \times S^5$ *[1–4]. For the operator family defined in Definition III.1,*

$$\mathcal{C}_{\mathrm{ex}} = \left\{ \mathcal{O}^a_\Delta(x),\ J^A_\mu(x),\ T_{\mu\nu}(x),\ \partial^\mu J^A_\mu(x) \right\}, \tag{30}$$

the following holds for any operator $\hat{\mathcal{O}} \in \mathcal{C}_{\mathrm{ex}}$*:*

$$\langle 0|\hat{\mathcal{O}}|0\rangle = 0\,. \tag{31}$$

More concretely,

$$\langle \mathcal{O}^a_\Delta(x)\rangle_0 = 0\,, \quad \langle J^A_\mu(x)\rangle_0 = 0\,, \quad \langle T_{\mu\nu}(x)\rangle_0 = 0\,, \quad \langle \partial^\mu J^A_\mu(x)\rangle_0 = 0\,. \tag{32}$$

Proof. The scalar sector follows from Propositions III.5 and III.15. The current sector follows from Propositions III.7, III.11, and III.16. The stress-tensor sector follows from Propositions III.9 and III.17. Combining these yields the claim. □

G. Remarks on scope

Remark III.19. The logical content of Theorem III.18 is that the 1-point function of each element of the operator family $\mathcal{C}_{\mathrm{ex}}$ vanishes in the source-free symmetric vacuum of the exact conformal theory. The correctness of this theorem does not depend on whether $\mathcal{C}_{\mathrm{ex}}$ contains "all operators necessary for nontrivial vacuum data."

Indeed, the proof of Theorem III.18 is given sector by sector using:

- the scale/conformal Ward identity for scalar primaries [13, 14],
- Lorentz invariance for vector currents [13, 14],
- Lorentz invariance and the trace Ward identity for the stress tensor [5, 13, 14],
- the conservation law for the current divergence [4, 6, 7].

Each argument acts individually on the adopted operator. Therefore, as long as an operator $\hat{\mathcal{O}}$ belongs to $\mathcal{C}_{\rm ex}$, the vanishing of its 1-point function is individually established.

What this paper does *not* show at this stage is the completeness of the operator family in the sense that "all operators necessary for nontrivial vacuum data are exhausted by $\mathcal{C}_{\rm ex}$." The unresolved question is:

$$\text{“All operators necessary for nontrivial vacuum data are exhausted by } \mathcal{C}_{\rm ex}\text{.”} \tag{33}$$

This is not an assumption needed for the proof of Theorem III.18, but an additional question that determines the ultimate scope of this work.

Consequently, what is rigorously established at this stage is

$$\forall\, \hat{\mathcal{O}} \in \mathcal{C}_{\rm ex}, \qquad \langle 0|\hat{\mathcal{O}}|0\rangle = 0\,, \tag{34}$$

while the proposition "any 1-point vacuum datum can be expressed as an element of $\mathcal{C}_{\rm ex}$" is deferred as a question for future extension.

H. What the theorem does not claim

The theorem of this section concerns the **vanishing of 1-point functions** and does not imply the triviality of the theory as a whole. In particular, the following are *not* consequences of the theorem:

1. Higher-point correlators vanish.
2. Vacuum fluctuations do not exist.
3. Nonlocal observables are trivial.
4. The same conclusion holds in excited or thermal states.

In fact, generically

$$\langle T_{00}(x)\, T_{00}(y)\rangle_0 \neq 0\,, \qquad \langle \mathcal{O}_\Delta(x)\, \mathcal{O}_\Delta(y)\rangle_0 \neq 0\,, \tag{35}$$

and the theory possesses a rich spectrum and nontrivial correlation structure [5, 13, 14]. What has been shown in this section is limited to the fact that, for each element of the

adopted operator family $\mathcal{C}_{\rm ex}$, **the symmetric vacuum itself carries no nonzero 1-point order parameter or density**.

Having established the vanishing, we turn in Section IV to the conditions under which deformations lift it.

IV. DEFORMATION $\Delta S_{\rm def}$ AND $\mathcal{C}_{\rm ex}$-RELATIVE NONZERO 1-POINT FUNCTIONS

In Section III, we showed that the 1-point function of each element of the adopted operator family $\mathcal{C}_{\rm ex}$ vanishes in the source-free symmetric vacuum of the exact conformal theory $S_{\rm CFT}$. As clarified in §III G, the correctness of this result does not depend on the completeness of $\mathcal{C}_{\rm ex}$. In this section, we systematize the types of deformations by which this $\mathcal{C}_{\rm ex}$-relative vanishing theorem is lifted.

The goal is not to leave the result $\langle\hat{\mathcal{O}}\rangle_0 = 0$ as a mere special case, but to organize **which assumption must be violated for which type of 1-point function in $\mathcal{C}_{\rm ex}$ to become nonzero**.

Throughout this section, the deformation from the exact conformal theory is written as

$$S = S_{\rm CFT} + \Delta S_{\rm def}\,, \tag{36}$$

where $\Delta S_{\rm def}$ violates at least one of the defining properties of the exact conformal theory introduced in §II C.

A. Necessary conditions for nonzero 1-point functions

As follows from the proof in Section III, the vanishing of the 1-point function for each element of $\mathcal{C}_{\rm ex}$ relied principally on four conditions: (1) translation invariance, (2) Lorentz invariance, (3) scale/conformal invariance, and (4) exact current conservation together with the source-free condition.

Proposition IV.1. *Let $\hat{\mathcal{O}} \in \mathcal{C}_{\rm ex}$. A necessary condition for $\langle\hat{\mathcal{O}}\rangle \neq 0$ is that at least one of the defining assumptions of the exact conformal theory is violated.*

Proof. Theorem III.18 gives $\langle\hat{\mathcal{O}}\rangle = 0$ as long as all defining assumptions hold. The contrapositive yields the claim. □

Remark IV.2. This is a necessary condition, not a sufficient one. For instance, introducing a relevant deformation may still leave certain 1-point functions vanishing, depending on residual symmetry or the choice of source. Below, we organize the specific circumstances under which each sector of $\mathcal{C}_{\rm ex}$ admits a nonzero 1-point function.

B. Relevant deformation and scalar 1-point function

The most basic deformation is to couple a relevant operator to the CFT. For a scalar primary $\mathcal{O}_\Delta$ with $\Delta < 4$,

$$\Delta S_{\rm rel} = \int d^4x \, \lambda \, \mathcal{O}_\Delta(x) \,, \tag{37}$$

where $[\lambda] = 4-\Delta > 0$, so that λ introduces an explicit scale into the theory. Scale/conformal symmetry is then lost, and the assumption of Proposition III.5 no longer holds [13, 14].

Proposition IV.3. *Under the deformation $S = S_{\rm CFT} + \int d^4x \, \lambda \, \mathcal{O}_\Delta(x)$ with $\Delta < 4$, scale invariance is explicitly broken, and the scalar 1-point function $\langle \mathcal{O}_\Delta(x) \rangle$ need no longer vanish.*

Proof. The relevant coupling λ carries mass dimension, providing a scale available in the vacuum. The scale Ward identity therefore no longer forces $\langle \mathcal{O}_\Delta \rangle = 0$ [13, 14]. By dimensional analysis, a quantity of the form $\lambda^{\Delta/(4-\Delta)}$ can be constructed, and when λ is the sole scale, the scalar 1-point function is expected to exhibit such a scaling. Hence, at least generically, the vanishing is no longer enforced. □

Remark IV.4. More precisely, $\langle \mathcal{O}_\Delta(x) \rangle = \delta W / \delta \lambda(x)$ [2, 3, 5, 9, 10]. If λ is regarded as a constant source, the nonvanishing of this 1-point function is tied to the λ-dependence of the vacuum energy.

C. Finite temperature and nonzero stress tensor

Introducing a finite temperature $T \neq 0$ is implemented by compactifying the Euclidean time: $\tau \sim \tau + \beta$, $\beta = T^{-1}$. This selects a thermal state rather than the vacuum, breaking both Lorentz invariance and scale invariance [4, 7].

Proposition IV.5. *In a thermal equilibrium state of $\mathcal{N} = 4$ SYM at $T \neq 0$, the stress-tensor sector of $\mathcal{C}_{\rm ex}$ satisfies $\langle T_{\mu\nu} \rangle_T \neq 0$ in general. In an isotropic thermal state,*

$$\langle T_{\mu\nu} \rangle_T = \mathrm{diag}(\varepsilon, p, p, p) \,, \tag{38}$$

with the conformal equation of state $\varepsilon = 3p$.

Proof. The temperature T introduces an explicit energy scale, and the thermal state is not a vacuum, so Lorentz invariance is lost. The assumption of Proposition III.9 therefore fails [13, 14]. Since $\mathcal{N} = 4$ SYM remains conformal, the thermal stress tensor is traceless but need not vanish [4, 7]. Isotropy and translation invariance constrain it to the above form. □

Remark IV.6. At strong coupling and large N, this thermal state is dual to an AdS-Schwarzschild black brane, whose holographic stress tensor gives $\varepsilon \propto N^2 T^4$, $p = \varepsilon/3$ [3, 4, 12].

D. Finite chemical potential and charge density

For a global symmetry current J^A_μ, introducing a time-component source defines a finite chemical potential:

$$\Delta S_\mu = \int d^4x \; \mu_A \, J^A_0(x) \, . \tag{39}$$

This permits a finite density in the vacuum and breaks Lorentz invariance.

Proposition IV.7. *With $\mu_A \neq 0$, the charge density in the current sector of $\mathcal{C}_{\text{ex}}$ can become nonzero: $\langle J^A_0(x) \rangle_\mu \neq 0$ in general.*

Proof. μ_A is a constant source for J^A_0, and $\langle J^A_0(x) \rangle_\mu = \delta W / \delta \mu_A(x)$ [2, 3, 5, 9, 10]. Finite μ_A singles out the time direction, breaking Lorentz invariance, so Proposition III.7 does not apply. Thermodynamically, $\langle J^A_0 \rangle$ equals the μ_A-derivative of the pressure or grand potential, and is therefore generically nonzero. □

Remark IV.8. On the AdS side, μ_A is implemented as the boundary value $A_t^{A(0)} = \mu_A$ of the bulk gauge field. The nonzero charge density corresponds to a nonzero normalizable coefficient $A_t^{A(2)}$ [9–11].

E. Vacuum selection and spontaneous symmetry breaking

While Section III fixed the conformal vacuum, the theory may possess multiple vacuum branches. $\mathcal{N} = 4$ SYM itself has a Coulomb branch admitting adjoint scalar VEVs [4, 6, 7].

Proposition IV.9. *In a Coulomb-branch vacuum of $\mathcal{N} = 4$ SYM, appropriate gauge-invariant operators in the scalar sector of $\mathcal{C}_{\text{ex}}$ can satisfy $\langle \mathcal{O}_\Delta(x) \rangle \neq 0$.*

Proof. On the Coulomb branch, the eigenvalues of the adjoint scalars Φ^I are nonzero, introducing an intrinsic length scale [4, 6, 7]. Conformal symmetry is spontaneously broken, so the assumption of Proposition III.5 fails. Gauge-invariant operators of the form $\text{Tr}(\Phi^{(I_1} \cdots \Phi^{I_k)})$ can then have generically nonzero VEVs. □

Remark IV.10. In AdS/CFT, this corresponds to a Coulomb-branch geometry departing from pure $\text{AdS}_5 \times S^5$ [1, 4, 7]. The nonvanishing of boundary 1-point functions is equivalent to normalizable deformations of bulk scalar modes, and chiral primary VEVs have been computed holographically [9–11, 45].

F. Explicit and anomalous current non-conservation

In Section III, exact global symmetry was assumed, giving $\langle \partial^\mu J^A_\mu \rangle_0 = 0$. When the current suffers an anomaly or explicit breaking, this conclusion is lifted.

Proposition IV.11. *If a current J_μ satisfies the anomaly equation*

$$\partial^\mu J_\mu = \mathcal{A}[\textit{background fields}]\,, \tag{40}$$

and if the background fields are configured so that $\mathcal{A} \neq 0$, then the divergence sector of $\mathcal{C}_{\text{ex}}$ satisfies $\langle \partial^\mu J_\mu \rangle = \langle \mathcal{A} \rangle \neq 0$.

Proof. Take the vacuum expectation value of the anomaly equation. □

Remark IV.12. It is important that $\partial^\mu J_\mu \neq 0$ (non-conservation) and $\langle J_0 \rangle \neq 0$ (finite density) are distinct conditions. To mimic baryogenesis-type phenomena, one typically needs not only an anomaly but also CP violation, departure from equilibrium, and a time-dependent background.

G. Time-dependent backgrounds

Time-dependent sources or curvature backgrounds destroy the static nature of the vacuum. In particular, in a cosmological background, the Poincaré invariance used on flat Minkowski space is lost.

Proposition IV.13. *Under a time-dependent background $g^{(0)}_{\mu\nu}(x)$ or time-dependent sources $J(x)$, $A_\mu(x)$, the $\mathcal{C}_{\rm ex}$-relative vanishing theorem of Section III does not hold in general. In particular, $\langle T_{\mu\nu}(x)\rangle$, $\langle \mathcal{O}_\Delta(x)\rangle$, and $\langle J_\mu(x)\rangle$ can become nonzero in a time-dependent manner.*

Proof. Propositions III.5–III.11 each assumed translation invariance, Lorentz invariance, or a static vacuum. In a time-dependent background these premises fail, and the vanishing theorem does not apply. □

Remark IV.14. This point will become essential in the comparison with $\Lambda > 0$ backgrounds, especially de Sitter-type geometries, where the horizon, thermal properties, and time-dependent coarse-graining do not permit the same structure of "complete vanishing of 1-point functions" seen in the source-free conformal vacuum of AdS [15–27].

H. Bulk image of the deformations

Each deformation described above is expressed on the AdS side as a deformation of the bulk geometry or bulk field profile [3, 4, 9–12]. The correspondence is summarized schematically as follows:

boundary deformation	bulk dual
$\lambda\,\mathcal{O}_\Delta$	$\phi_{(0)} \neq 0$ or scalar hair
$T \neq 0$	AdS black brane / black hole
$\mu_A \neq 0$	$A_t^{A(0)} \neq 0,\ A_t^{A(2)} \neq 0$
vacuum selection	normalizable moduli deformation
$g^{(0)}_{\mu\nu}(x) \neq \eta_{\mu\nu}$	nontrivial asymptotically AdS geometry

(41)

What these share is that the assumption of a "source-free, maximally symmetric pure-AdS geometry" used in Section III breaks down. As a result, the normalizable coefficients in the near-boundary expansion become nonzero, and the boundary 1-point functions can become nonzero [9–12].

Proposition IV.15. *If $\Delta S_{\rm def}$ induces a nontrivial classical solution departing from pure $\mathrm{AdS}_5 \times S^5$ on the bulk side, then the corresponding boundary 1-point functions of $\mathcal{C}_{\rm ex}$ can generically become nonzero.*

Proof. By holographic renormalization, boundary 1-point functions are given by normalizable coefficients of bulk fields [9–11]. In solutions departing from pure AdS, these coefficients are generically nonzero. □

I. General theorem

Collecting the individual cases above, the deformation principle of this paper is expressed as follows.

Theorem IV.16 (Nonzero realization)**.** *Consider the theory obtained by adding a deformation $\Delta S_{\rm def}$ to the exact conformal theory $S_{\rm CFT}$:*

$$S = S_{\rm CFT} + \Delta S_{\rm def}\,. \tag{42}$$

Suppose $\Delta S_{\rm def}$ violates at least one of the assumptions supporting the $\mathcal{C}_{\rm ex}$-relative vanishing theorem of Section III—Poincaré invariance, Lorentz invariance, scale/conformal invariance, the source-free condition, or exact current conservation. Then, for at least a part of $\mathcal{C}_{\rm ex}$, nonzero 1-point functions are permitted—the assumptions of the vanishing theorem no longer hold. For the finite-temperature, finite-density, Coulomb-branch, and anomalous deformations, constructive realizations are known; for time-dependent backgrounds, only the breakdown of the vanishing conditions is established here. More concretely, depending on the type of deformation, one or more of the following can occur:

$$\langle \mathcal{O}_\Delta \rangle \neq 0\,, \quad \langle J_\mu \rangle \neq 0\,, \quad \langle T_{\mu\nu} \rangle \neq 0\,, \quad \langle \partial^\mu J_\mu \rangle \neq 0\,. \tag{43}$$

Proof. Combine Propositions IV.3, IV.5, IV.7, IV.9, IV.11, IV.13, and IV.15. □

Remark IV.17. Like all results of this paper, Theorem IV.16 is $\mathcal{C}_{\rm ex}$-relative (see §II E and §III G).

J. Supplement: on sufficient conditions and genericity

Theorem IV.16 is not the strong claim that "if a deformation is introduced, then a specific 1-point function necessarily becomes nonzero." Which 1-point function becomes nonzero depends on the symmetry of the deformation, the boundary conditions, the choice of state, and the quantum numbers of the coupled operator.

A related distinction concerns the level of evidence. For finite temperature, finite density, Coulomb-branch, and anomalous deformations, the nonvanishing is established constructively—explicit formulas, thermodynamic identities, or known holographic computations provide concrete nonzero values. For time-dependent backgrounds, the argument shows only that the assumptions of the vanishing theorem fail; no explicit nonzero value is computed in this paper. The two levels—permission and constructive realization—are both recorded in the statement of Theorem IV.16.

However, the important structure is the converse direction. In a vacuum that is exactly conformal in the sense of Section III, all 1-point functions of $\mathcal{C}_{\rm ex}$ vanish; conversely, if a nonzero 1-point function for some element of $\mathcal{C}_{\rm ex}$ is observed, there must be a violation of the assumptions of the exact conformal theory behind it. In this sense, deformation is a necessary condition for $\mathcal{C}_{\rm ex}$-relative nontrivial vacuum data, and—at least within the scope of the adopted operator family—the emergence of nontrivial vacuum data is tied to the violation of the assumptions of the exact conformal theory.

We next apply this deformation framework to the $\Lambda > 0$ background in Section V.

V. CONNECTION TO THE $\Lambda > 0$ BACKGROUND

In Section III we showed that the 1-point function of every element of the adopted operator family $\mathcal{C}_{\rm ex}$ vanishes in the conformal vacuum of $\mathrm{AdS}_5 \times S^5$ and its boundary dual $\mathcal{N} = 4$ SYM. In Section IV we organized how this $\mathcal{C}_{\rm ex}$-relative vanishing theorem is lifted by the introduction of a deformation $\Delta S_{\rm def}$.

The purpose of this section is to apply that general framework to $\Lambda > 0$ backgrounds, in particular de Sitter spacetime. The aim is to clarify why the structure "source-free, maximally symmetric, symmetric vacuum" that held on $\mathrm{AdS}_5 \times S^5$ does not reproduce in the same sense on de Sitter geometry. More precisely, the issue is not that de Sitter admits "no vacuum at all," but that **the rigorous definition of observables via a boundary generating functional, the source/response dictionary, and the unambiguous treatment of the source-free vacuum that were used in AdS/CFT are not implemented in the same form for** $\Lambda > 0$ [15–21].

Below we first organize the difficulties in defining observables on de Sitter backgrounds, then describe the thermal properties arising from the Gibbons–Hawking temperature and

horizon. We subsequently confirm that $H \sim \sqrt{\Lambda}$ acts as an IR scale, and finally summarize, in the form of a theorem, how the assumptions supporting the $\mathcal{C}_{\rm ex}$-relative vanishing theorem of Section III are lost.

A. Observables in de Sitter spacetime

The d-dimensional de Sitter spacetime dS_d is the maximally symmetric solution with positive cosmological constant $\Lambda > 0$. Using the de Sitter radius $L_{\rm dS}$ or the Hubble rate H, it is characterized by [17, 20, 21]

$$R_{\mu\nu} = (d-1)\, H^2\, g_{\mu\nu}\,, \qquad \Lambda = \frac{(d-1)(d-2)}{2}\, H^2\,. \tag{44}$$

Although geometrically maximally symmetric, an essential difference from AdS appears when formulating quantum theory. The first concerns **the definition of observables**.

In AdS, a timelike conformal boundary exists at spatial infinity, on which boundary conditions, sources, and the generating functional can be rigorously defined [2, 3, 9–11]. The GKPW-type dictionary used in Section III,

$$W[J] = -S_{\rm ren}^{\rm on\text{-}shell}[\phi_{(0)} = J]\,, \tag{45}$$

relies on this structure. In de Sitter, the natural "boundaries" are the future infinity $\mathcal{I}^+$ and the past infinity $\mathcal{I}^-$, which differ in character from the timelike boundary of AdS (see FIG. 1) [16–21]. Furthermore, a local observer is confined to a causal patch by the cosmological horizon, so that the observable algebra of the full theory cannot be given purely in terms of asymptotic data in the same way as in AdS [16, 17, 19–21].

Proposition V.1. *In de Sitter spacetime, no timelike conformal boundary of the AdS type exists. Consequently, the source/response dictionary underlying the vanishing theorem of Section III cannot be applied in standard form.*

Proof. The GKPW dictionary of AdS is based on the specification of boundary conditions and sources on the timelike conformal boundary [2, 3]. In de Sitter, $\mathcal{I}^\pm$ are not timelike, and a local observer is confined to the interior of the horizon, so the structure in which a single global boundary functional yields all physical observables does not hold [16–21]. Hence the unambiguous formulation based on a boundary generating functional of the type used in Section III cannot be applied as it stands. □

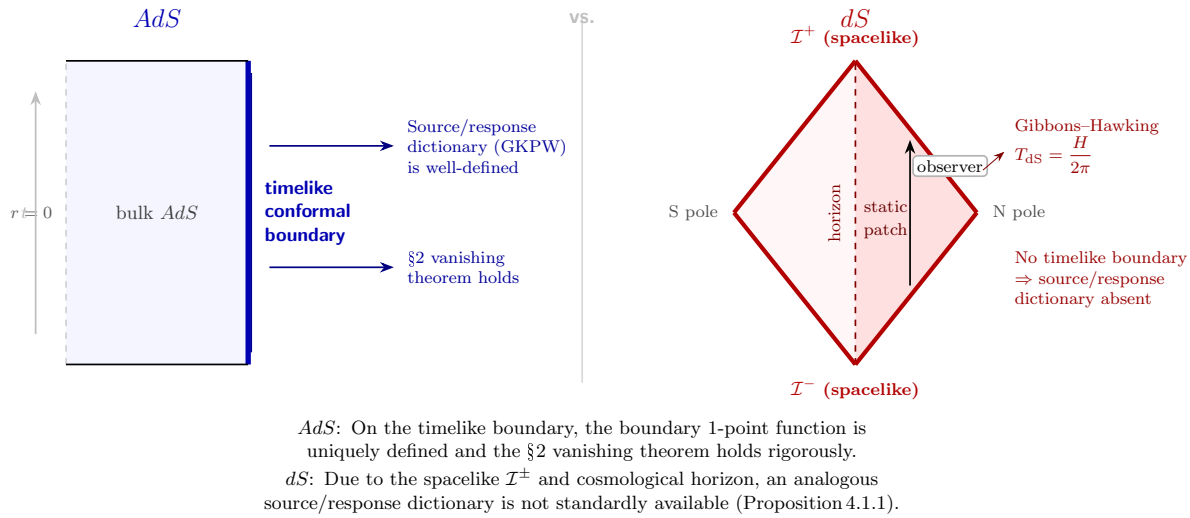

FIG. 1. Penrose diagrams for AdS (left) and de Sitter (right). In AdS, the timelike conformal boundary supports a well-defined source/response dictionary (GKPW), and the vanishing theorem of Section III holds rigorously. In de Sitter, the boundaries $\mathcal{I}^\pm$ are spacelike, and a local observer is confined to a static patch by the cosmological horizon at the Gibbons–Hawking temperature $T_{\rm dS} = H/2\pi$. The standard source/response dictionary is not available (Proposition V.1).

Remark V.2. This does not mean that all correlation functions or wavefunctions are undefined in de Sitter. In fact, the late-time wavefunction, in-in correlators, static patch observables, and dS/CFT-type proposals exist [18–21, 24, 25]. The claim here is the limited statement that the unambiguous source/response dictionary for boundary 1-point functions of the type supporting the rigorous theorem of Section III is not available in standard form.

B. Static patch and Gibbons–Hawking temperature

The second essential feature of de Sitter is that every geodesic observer possesses a cosmological horizon [15, 17, 20, 21]. In static patch coordinates the metric takes the form

$$ds^2 = -(1 - H^2 r^2)\, dt^2 + \frac{dr^2}{1 - H^2 r^2} + r^2\, d\Omega_{d-2}^2\,, \tag{46}$$

and the horizon is located at $r = H^{-1}$ [17, 20, 21]. Quantum-mechanically, this horizon is associated with a temperature [15]

$$T_{\rm dS} = \frac{H}{2\pi}\,. \tag{47}$$

Proposition V.3. *In the static patch of de Sitter spacetime, restricting the Bunch–Davies vacuum to a local observer yields a reduced state that behaves as a thermal density matrix at temperature* $T_{\rm dS} = H/(2\pi)$.

Proof. The Bunch–Davies vacuum is a de Sitter-invariant Gaussian state whose two-point function, through Euclidean continuation, possesses periodicity $\beta = 2\pi/H$ [22, 23]. Since the KMS condition with respect to static patch time is satisfied, the reduced density matrix for the local observer is a thermal state at temperature $T_{\rm dS} = H/(2\pi)$ [15, 17, 20, 21, 23]. □

Corollary V.4. *A local observer in de Sitter does not observe an "empty" pure vacuum in general, but a thermally coarse-grained state.*

Remark V.5. This fact does not directly negate the structure used in Section III—"vanishing of the 1-point function in the source-free, Poincaré-invariant, symmetric vacuum"—but it drastically changes its **physical interpretation**. Whereas in AdS the source-free vacuum can be treated even locally as a cold reference state, in de Sitter the same notion of "cold emptiness" cannot be assigned to the static patch observer [15–17, 20, 21, 23].

C. Introduction of an IR scale by H

At the core of the scalar 1-point function vanishing theorem in Section III was the absence of any intrinsic scale in the conformal vacuum [13, 14]. In de Sitter, even if the UV side of the theory is scale-invariant, the background curvature itself introduces a length scale [17, 20, 21]

$$H^{-1}\,. \tag{48}$$

Proposition V.6. *In quantum field theory on a de Sitter background, the Hubble rate H acts as an external IR scale, so that the argument for the vanishing of the 1-point function based on the scale Ward identity in flat-space CFT does not hold in general.*

Proof. In a flat-space CFT, $\langle \mathcal{O}_\Delta \rangle = 0$ follows from the absence of any scale that could build a quantity of dimension $\Delta > 0$ [13, 14]. On a de Sitter background, the curvature radius H^{-1} is given as background data, so by dimensional analysis a quantity

$$\langle \mathcal{O}_\Delta \rangle \sim c\, H^{\Delta} \tag{49}$$

can be constructed [17, 20, 21]. The vanishing proof based on scale invariance in Section III therefore does not apply. □

Remark V.7. Whether $\langle \mathcal{O}_\Delta \rangle$ actually becomes nonzero depends on the quantum numbers of the operator, the renormalization scheme, the choice of state, and the way de Sitter symmetry is realized [20–23, 26, 27]. What matters here is that **the enforcing power that requires it to vanish is lost**.

D. Mismatch between de Sitter symmetry and the local observer

Since de Sitter spacetime is maximally symmetric as a whole, one might object that "in a symmetric vacuum the 1-point function should still vanish." However, the key point is that the **global dS-invariant state** and the **physical state actually accessible to the static patch observer** are not identical [17, 20, 21].

Even though the global Bunch–Davies state is de Sitter-invariant, its restriction to the static patch is a thermal mixed state [15, 17, 20, 21, 23]. The quantity relevant to a local observer is therefore not the global vacuum expectation value but the expectation value with respect to the patch-restricted density matrix.

Proposition V.8. *In de Sitter, global de Sitter invariance alone does not guarantee the vanishing of 1-point functions for a local observer.*

Proof. Global de Sitter invariance refers only to the symmetry of the total state; the accessible algebra of a local observer is restricted to the causal patch [17, 20, 21]. This restriction maps the global pure state to a local mixed state. For a mixed thermal state, the vanishing theorem of Section III, which was established for the Lorentz-invariant vacuum, cannot be applied as it stands [15, 17, 20, 21, 23]. □

Remark V.9. For example, in the static patch thermal state the local expectation value of the stress-energy tensor is defined in an observer-dependent manner, and the renormalized stress tensor can become nonzero. At a minimum, the argument of Section III—concluding uniquely 0 from symmetry alone—breaks down [17, 20, 21].

E. In-in formalism and late-time observables

In a cosmological background, the natural observables are often defined not via an AdS-type boundary S-matrix or Euclidean generating functional but via the Schwinger–Keldysh

in-in formalism [24, 25]. That is, for an initial state $|\Psi_{\rm in}\rangle$, the expectation value at time t is

$$\langle \mathcal{O}(t)\rangle = \langle \Psi_{\rm in}|\, \bar{T} \exp\Big(i\int_{t_0}^{t} H_I(t')\, dt'\Big)\, \mathcal{O}_I(t)\, T \exp\Big(-i\int_{t_0}^{t} H_I(t')\, dt'\Big)|\Psi_{\rm in}\rangle\,, \tag{50}$$

where T and $\bar{T}$ denote time-ordering and anti-time-ordering respectively [24, 25]. In this framework, the dynamics of a time-dependent background and horizon are directly reflected in the observables.

Proposition V.10. *In a de Sitter-type cosmological background, the natural observables are given by in-in expectation values or late-time wavefunction coefficients rather than by boundary 1-point functions.*

Proof. In de Sitter, asymptotic scattering states or timelike boundary sources of the AdS type are not available in general, while cosmological observables are defined as expectation values at finite times or at late times [19–21]. These are described by the Schwinger–Keldysh formalism or the wavefunction of the universe [24, 25]. □

Remark V.11. This point clearly separates the "vanishing of the 1-point function in a source-free CFT vacuum" of Section III from the cosmologically relevant late-time observables such as $\langle \delta\rho/\rho\rangle$ and $\langle \zeta\zeta\rangle$ [24, 25]. In de Sitter, the nontriviality of the vacuum manifests not only at the level of 1-point functions but also through effective classicalization via mode freezing and decoherence.

F. $\Lambda > 0$ viewed as a deformation

In light of the general framework of Section IV, the $\Lambda > 0$ background can be regarded as a type of geometric deformation $\Delta S_{\rm def}$ of the exact conformal theory $S_{\rm CFT}$. What this deformation breaks is principally [15–21]:

1. flat-space Poincaré invariance,
2. the definition of observables based on a global timelike boundary,
3. the interpretation of the local state as a cold vacuum,
4. scalelessness.

Definition V.12. *Let the geometric deformation due to a positive cosmological constant be denoted by*

$$\Delta S_{\Lambda>0}\,. \tag{51}$$

This is regarded as a deformation that breaks the assumptions of the exact conformal theory not only in the matter sector but also in the definition of observables, the local interpretation of the vacuum, and the IR structure.

Proposition V.13. $\Delta S_{\Lambda>0}$ *breaks at least the following assumptions supporting the* $\mathcal{C}_{\mathrm{ex}}$*-relative vanishing theorem of Section III: the flat-space implementation of scale invariance, the source/response dictionary based on a timelike boundary, and the cold-vacuum interpretation for a local observer.*

Proof. The flat-space implementation of scale invariance is broken by the introduction of the Hubble scale H [17, 20, 21]. The source/response dictionary based on a timelike boundary is replaced by the causal structure of de Sitter [16–21]. The vacuum as seen by a local observer appears as a thermal state with the Gibbons–Hawking temperature, so the cold-vacuum interpretation is lost [15, 17, 20, 21, 23]. □

G. Main result: AdS-type exact conformal invariance is not stable in de Sitter

Combining all of the above, the main claim of this section is summarized as follows.

Proposition V.14. *The* $\mathcal{C}_{\mathrm{ex}}$*-relative vanishing theorem for 1-point functions in the source-free, symmetric vacuum established in Section III for* $\mathrm{AdS}_5 \times S^5/\mathcal{N}=4$ *SYM does not hold in the same form on a* $\Lambda>0$ *background in general. More specifically, on a de Sitter-type background:*

1. *the AdS/CFT-type timelike boundary source/response dictionary is not available in standard form,*

2. *for a static patch observer, the vacuum appears as a thermal state at temperature* $T_{\mathrm{dS}} = H/2\pi$*,*

3. *the background curvature* H *introduces an IR scale, and*

4. *cosmological observables are defined by in-in expectation values or the late-time wavefunction,*

so that the assumptions supporting the vanishing theorem of Section III are not preserved. Consequently, the $\Lambda > 0$ background is understood as a form of the deformation ΔS_{def} in the sense of Section IV.

Proof. Follows by combining Propositions V.1, V.3, V.6, V.8, V.10, and V.13. □

Remark V.15. Unlike Theorem III.18, which is deduced from Ward identities and holographic renormalization, the above proposition is a structural observation about the failure of assumptions on a $\Lambda > 0$ background. It does not compute a specific nonzero 1-point function on de Sitter; it identifies why the vanishing theorem cannot be applied.

H. Physical implications

Proposition V.14 does not assert that "in de Sitter, all 1-point functions of $\mathcal{C}_{\mathrm{ex}}$ necessarily become nonzero." What it asserts is that the framework of the exact conformal theory that held on $\mathrm{AdS}_5 \times S^5$ becomes structurally unstable for $\Lambda > 0$. This instability is consistent with the emergence of nontrivial observables in cosmological backgrounds [15–27].

In particular, in inflationary or quasi-de Sitter backgrounds, quantum fluctuations freeze after horizon crossing and become the seeds of late-time density perturbations [24, 25]. What is essential there is not the 1-point function that vanished in Section III but **the IR scale** H, the **time-dependent background**, the **coarse-graining by the horizon**, and the **in-in observable** framework [15–27]. In a de Sitter-type universe it is more natural to regard the background geometry itself as already implementing the deformation, rather than to take the "cold symmetric vacuum of an exact conformal theory" as the starting point.

In Section VI we collect the results of Sections III–V into the unification theorem.

VI. UNIFICATION THEOREM

In Section III we showed that the 1-point function of every element of the adopted operator family $\mathcal{C}_{\mathrm{ex}}$ vanishes in the source-free, symmetric vacuum of the exact conformal theory S_{CFT}. In Section IV we organized how this $\mathcal{C}_{\mathrm{ex}}$-relative vanishing theorem is lifted

by the introduction of a deformation $\Delta S_{\rm def}$. In Section V we showed that, on a $\Lambda > 0$ background—in particular de Sitter spacetime—the source-free, symmetric vacuum structure of $\mathrm{AdS}_5 \times S^5$ is not stably realized in the same sense [1–27].

The purpose of this section is to assemble these results into a single unified proposition. It is essential to separate clearly the part that holds as a theorem from the physical interpretation placed on top of it. What holds as a theorem is the structure that, **relative to the adopted operator family $\mathcal{C}_{\rm ex}$, all 1-point functions vanish in the symmetric vacuum of the exact conformal theory, and their nonvanishing is permitted once a deformation is introduced**.

A. Unification theorem

We collect the notation introduced in the preceding sections. Write the exact conformal theory as $S_{\rm CFT}$, with the canonical example being $\mathcal{N} = 4$ $SU(N)$ SYM on $\mathbb{R}^{1,3}$ and its holographic dual, source-free type IIB string theory on $\mathrm{AdS}_5 \times S^5$ [1–4]. The operator family of candidate 1-point observables is

$$\mathcal{C}_{\rm ex} = \left\{ \mathcal{O}^a_\Delta(x),\ J^A_\mu(x),\ T_{\mu\nu}(x),\ \partial^\mu J^A_\mu(x) \right\}, \tag{52}$$

where $\mathcal{O}^a_\Delta$ is a scalar primary with $\Delta > 0$, J^A_μ is an exact global symmetry current, and $T_{\mu\nu}$ is the stress tensor [4–7, 13, 14].

Theorem VI.1. *In the source-free, symmetric vacuum $|0\rangle$ of the exact conformal theory $S_{\rm CFT}$, for every $\hat{\mathcal{O}} \in \mathcal{C}_{\rm ex}$,*

$$\langle 0|\hat{\mathcal{O}}|0\rangle = 0\,. \tag{53}$$

On the other hand, in the theory

$$S = S_{\rm CFT} + \Delta S_{\rm def}\,, \tag{54}$$

if $\Delta S_{\rm def}$ breaks at least one of the assumptions supporting the vanishing theorem of Section III, then for at least part of $\mathcal{C}_{\rm ex}$,

$$\langle \hat{\mathcal{O}} \rangle_S \neq 0 \tag{55}$$

is permitted; for the specific deformations listed in Theorem IV.16, constructive realizations are known. Furthermore, the $\Lambda > 0$ background can be regarded as a geometric implementa-

tion of such $\Delta S_{\rm def}$ through the definition of observables, the thermal properties of the local vacuum, and the introduction of the IR scale H [15–27].

Proof. The first claim follows from Theorem III.18. The second from Theorem IV.16. The third from Proposition V.14. □

Remark VI.2. Theorem VI.1 is $\mathcal{C}_{\rm ex}$-relative (see §II E and §III G).

B. $\mathcal{C}_{\rm ex}$-relative definition of 1-point vacuum data

The theorems of this paper are meaningful when 1-point vacuum data are adopted as the class of 1-point functions belonging to $\mathcal{C}_{\rm ex}$. Under this definition, 1-point vacuum data are nontrivial physical quantities carried by the vacuum or state itself, and their presence or absence is determined by the nonvanishing of the expectation value [2, 3, 5, 9, 10].

Definition VI.3. *A state ρ is said to possess $\mathcal{C}_{\rm ex}$**-relative nontrivial vacuum data** with respect to $\hat{\mathcal{O}} \in \mathcal{C}_{\rm ex}$ if*

$$\langle \hat{\mathcal{O}} \rangle_\rho \neq 0 \,. \tag{56}$$

Remark VI.4. This definition differs from the presence or absence of particle content, of higher-point correlation functions, or of vacuum fluctuations [5, 13, 14]. The claim of this paper thus rests on the operational definition

$$\mathcal{C}_{\rm ex}\text{-relative nontrivial vacuum data} \quad \Longleftrightarrow \quad \text{nonzero 1-point function of some } \hat{\mathcal{O}} \in \mathcal{C}_{\rm ex} \,. \tag{57}$$

Remark VI.5. Here $\mathcal{C}_{\rm ex}$ is a natural operator family providing candidate 1-point observables, but its completeness—i.e., whether all operators necessary for nontrivial vacuum data are exhausted by this family—is not assumed in this paper. What is rigorously shown is the fact that, for every operator belonging to $\mathcal{C}_{\rm ex}$, the 1-point function vanishes in the symmetric vacuum of the exact conformal theory.

C. Corollary: an exact conformal theory carries no $\mathcal{C}_{\rm ex}$-relative nontrivial vacuum data

Corollary VI.6. *The source-free, symmetric vacuum of the exact conformal theory $S_{\rm CFT}$ carries no nontrivial vacuum data in the sense of the operator family $\mathcal{C}_{\rm ex}$ adopted in this*

paper.

Proof. By Theorem VI.1, all 1-point functions of $\mathcal{C}_{\text{ex}}$ vanish. □

Remark VI.7. This corollary does not mean that the exact conformal theory is physically trivial. The theory still possesses a nontrivial spectrum, higher-point correlation functions, Wilson loops, scattering amplitudes, and a holographic dual geometry [1–7, 9–14]. What is claimed is merely that, **with respect to the quantities belonging to the adopted operator family $\mathcal{C}_{\text{ex}}$, the vacuum itself carries no nonzero 1-point order parameter or density**.

D. Corollary: an observed nonzero quantity implies breakdown of exact conformal invariance

Reading Theorem VI.1 in the converse direction, an observed nonzero 1-point quantity for some operator in $\mathcal{C}_{\text{ex}}$ implies that at least one of the assumptions of the exact conformal theory is violated.

Corollary VI.8. *If, in a state ρ, some $\hat{\mathcal{O}} \in \mathcal{C}_{\text{ex}}$ satisfies*

$$\langle \hat{\mathcal{O}} \rangle_\rho \neq 0 \,, \tag{58}$$

then ρ is not the source-free, symmetric vacuum of the exact conformal theory assumed in Section III. More precisely, at least one of the following is violated: Poincaré invariance, Lorentz invariance, scale/conformal invariance, the source-free condition, or exact current conservation.

Proof. This is the contrapositive of Theorem III.18. □

Remark VI.9. This corollary constitutes the physical core of the vanishing theorem and the deformation principle of this paper. The "deformation" referred to here does not mean a mathematical inconsistency or theoretical pathology. What it means is that, **at least within the scope of $\mathcal{C}_{\text{ex}}$, the vacuum does not fully preserve symmetry, scalelessness, and the source-free condition**.

E. Position of the $\Lambda > 0$ background

Incorporating the results of Section V, a positive cosmological constant implements the departure from exact conformal invariance in a manner different from a matter-sector Lagrangian deformation: it is a geometric deformation acting through the background geometry, the definition of observables, and the thermal nature of the local state [15–27].

Proposition VI.10. *A $\Lambda > 0$ background does not remain within exact conformal invariance in the flat-space conformal vacuum sense. Hence a cosmological background with $\Lambda > 0$ can be regarded as a geometric ΔS_{def} in the sense of this paper.*

Proof. By Proposition V.14, the assumptions supporting the $\mathcal{C}_{\mathrm{ex}}$-relative vanishing theorem of Section III are not preserved on a de Sitter background. □

Remark VI.11. This proposition does not assert that "in de Sitter a specific 1-point function of $\mathcal{C}_{\mathrm{ex}}$ necessarily becomes nonzero." What it asserts is the structural statement that the framework of the exact conformal theory that held on $\mathrm{AdS}_5 \times S^5$ cannot be carried over as it stands to a $\Lambda > 0$ background [15–21].

F. Final formulation

In light of the above, the core proposition of this paper can be summarized in the following concise form.

Definition VI.12. *For the adopted operator family $\mathcal{C}_{\mathrm{ex}}$, define the $\mathcal{C}_{\mathrm{ex}}$-**relative vacuum data** of a state ρ by*

$$\mathfrak{E}_{\mathcal{C}_{\mathrm{ex}}}(\rho) = \left\{ \langle \hat{\mathcal{O}} \rangle_\rho \;\middle|\; \hat{\mathcal{O}} \in \mathcal{C}_{\mathrm{ex}} \right\}. \tag{59}$$

Hereafter, $\rho_0 := |0\rangle\langle 0|$ denotes the density matrix of the source-free, symmetric vacuum.

Theorem VI.13. *For the source-free, symmetric vacuum ρ_0 of the exact conformal theory,*

$$\mathfrak{E}_{\mathcal{C}_{\mathrm{ex}}}(\rho_0) = \{0\}. \tag{60}$$

On the other hand, for a state ρ_{def} obtained by a deformation $\Delta S_{\mathrm{def}} \neq 0$, in general

$$\mathfrak{E}_{\mathcal{C}_{\mathrm{ex}}}(\rho_{\mathrm{def}}) \neq \{0\} \tag{61}$$

is permitted.

Proof. The former is the first part of Theorem VI.1, and the latter is its second part. □

Remark VI.14. This theorem is $\mathcal{C}_{\rm ex}$-relative (see §II E and §III G).

G. Physical interpretation

Here, for the first time, we can give a minimal physical interpretation to the rigorous results obtained so far. The exact conformal theory $S_{\rm CFT}$ is mathematically closed and is the theory most tightly controlled by symmetry. Yet, precisely because of that exact conformal invariance, its symmetric vacuum carries—at least within the scope of $\mathcal{C}_{\rm ex}$—no nonzero 1-point order parameter, no nonzero density, and no nonzero production rate. Therefore, $\mathcal{C}_{\rm ex}$-relative nontrivial vacuum data emerge not from the exact conformal theory itself but only through its breaking, deformation, or vacuum selection.

This interpretation can be written in formulae; the content of this paper is summarized in the following schema:

$$S_{\rm CFT} \quad \Longrightarrow \quad \forall\, \hat{\mathcal{O}} \in \mathcal{C}_{\rm ex},\ \langle \hat{\mathcal{O}} \rangle = 0\,, \tag{62}$$

$$S_{\rm CFT} + \Delta S_{\rm def} \quad \Longrightarrow \quad \exists\, \hat{\mathcal{O}} \in \mathcal{C}_{\rm ex} \text{ such that } \langle \hat{\mathcal{O}} \rangle \neq 0 \text{ is allowed.} \tag{63}$$

In this sense, nontrivial vacuum data in the sense of this paper are understood not as the mere well-definedness of the theory but as the fact that, **with respect to the adopted operator family $\mathcal{C}_{\rm ex}$, the state carries nontrivial 1-point data not enforced by symmetry**.

H. Limitations and open problems

The discussion in this paper has a clear domain of applicability.

First, the 1-point vacuum data treated here are limited to 1-point functions belonging to $\mathcal{C}_{\rm ex}$. Since much of cosmological structure formation and classicalization is described through 2-point functions, non-Gaussianity, decoherence, and open-system dynamics, an extension is needed to incorporate them into the present framework [24, 25].

Second, the discussion of the $\Lambda > 0$ background does not presuppose a rigorous duality of the AdS/CFT type; it is a structural argument based on the standard difficulties in defining

observables in de Sitter and on horizon thermality [15–21]. The results of Section V should therefore be understood not as rigorous holographic theorems in the sense of Section III but as a **systematic organization of the reasons for the breakdown of exact conformal invariance**.

Third, a more concrete dictionary is needed between the abstract operator family $\mathcal{C}_{\rm ex}$ and Standard Model phenomena such as baryon asymmetry, electroweak symmetry breaking, and cosmological perturbations. This is a task to be refined in the context of top-down compactifications, the introduction of flavor branes, or holographic effective field theory [4, 7, 8].

Fourth, the most essential open problem is the completeness of $\mathcal{C}_{\rm ex}$: whether all operators necessary for nontrivial vacuum data are exhausted by $\mathcal{C}_{\rm ex}$ is unproven in this paper. Consequently, the conclusion rigorously established at the present stage is $\mathcal{C}_{\rm ex}$-relative.

I. Summary of this section

In this paper, $\mathrm{AdS}_5 \times S^5/\mathcal{N} = 4$ SYM was adopted as the canonical example of an exact conformal theory, and we showed that the 1-point function of every element of the operator family $\mathcal{C}_{\rm ex}$ defined there vanishes in the source-free, symmetric vacuum [1–14]. Next, we systematized that these 1-point functions belonging to $\mathcal{C}_{\rm ex}$ can become nonzero once deformations $\Delta S_{\rm def}$ such as relevant deformations, finite temperature, finite density, vacuum selection, and time-dependent backgrounds are introduced [3, 4, 9–12]. Furthermore, we showed that a $\Lambda > 0$ background can be regarded as a geometric implementation of such a deformation through the definition of observables, the thermal horizon, and the introduction of an IR scale [15–27].

The final conclusion of this paper is therefore:

$$\boxed{\text{The symmetric vacuum of the exact conformal theory carries no } \mathcal{C}_{\rm ex}\text{-relative nontrivial 1-point data.}} \tag{64}$$

$$\boxed{\text{The emergence of } \mathcal{C}_{\rm ex}\text{-relative nontrivial 1-point data requires a deformation } \Delta S_{\rm def}.} \tag{65}$$

This claim is rigorously formulated on $\mathrm{AdS}_5 \times S^5$ insofar as 1-point vacuum data are adopted as the class of 1-point functions belonging to $\mathcal{C}_{\rm ex}$. A cosmological background

with $\Lambda > 0$ is then positioned as an example in which that exact conformal invariance is geometrically broken [15–27].

At the same time, this paper does not claim that the totality of operators necessary for nontrivial vacuum data is exhausted by $\mathcal{C}_{\mathrm{ex}}$. This paper should therefore be understood not as the final form of a completed classification but as a **rigorous first step toward the vanishing theorem and the deformation principle, with the $\mathcal{C}_{\mathrm{ex}}$-relative vanishing theorem at its core**.

In the following sections we address the open problem of the completeness of $\mathcal{C}_{\mathrm{ex}}$ left in this section, introducing the "four-layer structure of vacuum data" that includes observables not belonging to $\mathcal{C}_{\mathrm{ex}}$—nonlocal operators, information-theoretic quantities, and higher-point correlation functions (Section VII). We then unify this four-layer classification with the deformation theory developed up to this section, systematically describing how each layer responds under deformation (Section VIII).

VII. GENERALIZED OBSERVABLES AND THE SCOPE OF THE 1-POINT FRAMEWORK

In Section VI we stated the unification theorem relative to the adopted operator family $\mathcal{C}_{\mathrm{ex}}$ and showed that all 1-point functions of $\mathcal{C}_{\mathrm{ex}}$ vanish in the source-free, symmetric vacuum of the exact conformal theory. At the same time, in §VI H the completeness of $\mathcal{C}_{\mathrm{ex}}$—whether all operators necessary for nontrivial vacuum data are exhausted by this family—was left as an open problem.

The purpose of this section is to address that open problem directly. Concretely, we examine observables not belonging to $\mathcal{C}_{\mathrm{ex}}$—nonlocal operators, information-theoretic quantities, and higher-point correlation functions—and rigorously determine whether the vanishing theorem of Section III holds for them.

The main result is as follows: the vanishing theorem of Section III is **a result specific to local, 1-point, intrinsic order parameters** and cannot in general be extended to nonlocal, relational, or information-theoretic quantities. This fact does not indicate a weakness of the theorem but, on the contrary, sharply characterizes the nature of the 1-point vacuum data that the vanishing theorem captures.

A. Setup: conditions for the vanishing theorem to hold

We recall the proof structure on which the vanishing theorem (Theorem III.18) relies. As appeared typically in Proposition III.5, the proof for a scalar primary $\mathcal{O}_\Delta(x)$ proceeded in three steps:

1. Translation invariance: $\langle \mathcal{O}_\Delta(x) \rangle = c_0$ (constant).
2. Scale invariance: $c_0 = t^{-\Delta}\, c_0$ for all $t > 0$.
3. $\Delta > 0$ forces $c_0 = 0$.

For this argument to apply, the observable must satisfy the following three conditions:

$$\boxed{\begin{cases} \text{(a) Linear in the state (expressible as an operator expectation value),} \\ \text{(b) Requires no auxiliary geometric data (no contour, region, or reference state),} \\ \text{(c) Defined as a single-point local operator with positive scaling dimension } \Delta > 0. \end{cases}} \tag{66}$$

These three conditions abstract the conditions for the scalar sector appearing typically in Proposition III.5. For Theorem III.18 as a whole, additional Ward identities—Lorentz invariance, the trace Ward identity, current conservation—corresponding to the Lorentz quantum numbers of each operator are also required.

Below we examine, in turn, observables that violate each of these conditions:

- the defect sector (§VII B) violates conditions (b) and (c),
- the information-theoretic sector (§VII C) violates conditions (a) and (b),
- the correlation sector (§VII D) violates conditions (b) and (c).

B. Defect sector: Wilson loop

a. Definition. In $\mathcal{N} = 4$ SYM, the 1/2-BPS Wilson loop is defined by [28–31]

$$W(C) = \frac{1}{N} \operatorname{Tr} \mathcal{P} \exp \oint_C \big(i\, A_\mu\, \dot{x}^\mu + \Phi^I n^I\, |\dot{x}| \big)\, d\tau\,, \tag{67}$$

where C is a closed contour and n^I is a unit vector on S^5. Below we work with the 1/2-BPS circular Wilson loop on Euclidean $\mathbb{R}^4$ obtained by Wick rotation. Our subject is the circular

Wilson loop whose renormalized expectation value is finite due to 1/2-BPS protection and can be computed exactly via localization [29–31].

b. Exact result at finite N and the planar limit. By Pestun's supersymmetric localization [31], the expectation value of the 1/2-BPS circular Wilson loop is computed exactly by a Gaussian matrix model. In the standard $U(N)$ theory [29–31],

$$\langle W_\circ \rangle_{U(N)} = \frac{1}{N}\, L^1_{N-1}\Big(-\frac{\lambda}{4N}\Big)\exp\Big(\frac{\lambda}{8N}\Big)\,, \tag{68}$$

where L^α_n is a generalized Laguerre polynomial and $\lambda = g^2_{\rm YM} N$ is the 't Hooft coupling.

In the planar large-N limit [29, 30],

$$\langle W_\circ \rangle = \frac{2}{\sqrt{\lambda}}\, I_1(\sqrt{\lambda}) + O(N^{-2})\,, \tag{69}$$

where I_1 is the modified Bessel function of the first kind. Below we use this planar formula.

Remark VII.1. The exact formula above is written for $U(N)$ in standard form, but in the planar limit the difference between $SU(N)$ and $U(N)$ is $O(N^{-2})$, consistent with the $\mathcal{N} = 4$ $SU(N)$ SYM discussion of this paper [29–31]. What is needed in this section is the nonvanishing in the planar limit, and the discussion proceeds within that scope.

c. Weak-coupling and strong-coupling behavior. In the weak-coupling expansion [29, 30],

$$\langle W_\circ \rangle = 1 + \frac{\lambda}{8} + O(\lambda^2)\,, \tag{70}$$

and in the strong-coupling regime [29, 30],

$$\langle W_\circ \rangle \sim \sqrt{\frac{2}{\pi}}\, \lambda^{-3/4}\, e^{\sqrt{\lambda}}\,. \tag{71}$$

In particular, $\log\langle W_\circ \rangle \sim \sqrt{\lambda}$, which is consistent on the AdS side with the renormalized area of the fundamental-string minimal worldsheet, $S_{\rm ren} = -\sqrt{\lambda}$ and $\langle W_\circ \rangle \sim e^{-S_{\rm ren}}$ [28]. Numerical consistency in both the weak-coupling and strong-coupling limits is confirmed in [Comp. 1].

Proposition VII.2. *In the conformal vacuum of $\mathcal{N} = 4$ SYM, the planar expectation value of the 1/2-BPS circular Wilson loop satisfies*

$$\langle W_\circ \rangle_0 = \frac{2}{\sqrt{\lambda}}\, I_1(\sqrt{\lambda}) > 0 \qquad (\forall\, \lambda > 0)\,. \tag{72}$$

Therefore, if $W(C)$ were added to $\mathcal{C}_{\rm ex}$, the vanishing theorem of Section III (Theorem III.18) would no longer hold.

Proof. $I_1(z) > 0$ for all $z > 0$ is a basic property of the modified Bessel function. Since $\lambda > 0$ implies $\sqrt{\lambda} > 0$, we have $I_1(\sqrt{\lambda}) > 0$ and hence $\langle W_\circ \rangle_0 = (2/\sqrt{\lambda})\, I_1(\sqrt{\lambda}) > 0$. □

d. Structural reason why the vanishing theorem does not apply to the Wilson loop. If one tries to apply the three-step argument of Proposition III.5 to the Wilson loop, it fails at every step:

Step 1 (translation). $W(C)$ depends on the contour C, not on a point x. Translation invariance gives only $\langle W(C) \rangle = \langle W(C + a) \rangle$; "reducing to a constant" does not occur. $\langle W \rangle$ is a functional of C. Conditions (b) and (c) of (66) are violated here.

Step 2 (scaling). Under $x \mapsto tx$, $C \mapsto tC$, but conformal transformations map any circle to any circle, so $\langle W(C_R) \rangle$ is independent of the radius R. Scaling tells us that $\langle W \rangle$ is a function of λ and N only; it does **not force it to vanish**.

Step 3 ($\Delta > 0$). The Wilson loop is a dimensionless quantity and does not possess a positive scaling dimension $\Delta > 0$ in the sense of Section III. Hence a constraint of the type $c_0 = t^{-\Delta} c_0$ does not arise.

Thus the vanishing argument fails at all three steps for the Wilson loop.

e. Physical interpretation: intrinsic 1-point data versus probe response. The 1-point functions of $\mathcal{C}_{\text{ex}}$ whose vanishing was shown in Section III are quantities requiring no additional geometric data. They measure **the intrinsic 1-point data that the vacuum itself carries**, and they vanish in the symmetric vacuum because such data are absent.

The Wilson loop, on the other hand, measures the response of the theory to a **geometric probe supplied from outside**—the contour C. $\langle W(C) \rangle \neq 0$ does not mean that "the vacuum has chosen a direction" but that "the theory has a nontrivial capacity to respond to a probe." This response is determined solely by the defining data of the theory (λ, N) and the geometry of the probe (the shape of C).

Definition VII.3.

- ***Intrinsic 1-point data:*** *quantities that ask whether the vacuum carries a nonzero expectation value without any additional geometric data. $\mathcal{C}_{\text{ex}}$ corresponds to this class.*

- ***Probe-response data (defect type):*** *quantities that measure the response of the theory after supplying a geometric probe (a contour C, a surface Σ, etc.). Wilson loops, 't Hooft loops, surface operators, etc. correspond to this class.*

Proposition VII.4. *The source-free, symmetric vacuum of the exact conformal theory satisfies the vanishing theorem for intrinsic 1-point data ($\mathcal{C}_{\text{ex}}$), but can have a nonzero response for probe-response data.*

Proof. The former follows from Theorem III.18. The latter follows from Proposition VII.2, which shows that $W_\circ$ is nonzero in the symmetric vacuum of the exact conformal theory. □

C. Information-theoretic sector

a. Entanglement entropy is not an operator expectation value. The entanglement entropy for a spatial region A is defined by

$$S_A(\rho) = -\mathrm{Tr}(\rho_A \log \rho_A)\,. \tag{73}$$

If a fixed operator $\hat{S}_A$ existed such that $S_A(\rho) = \mathrm{Tr}(\rho\,\hat{S}_A)$ for all states ρ, the right-hand side would be linear in ρ. However, $S_A(\rho)$ is a nonlinear (concave) function of ρ, so no such fixed operator exists in general. This corresponds to the violation of condition (a) in (66).

Proposition VII.5. *The entanglement entropy $S_A(\rho)$ cannot in general be expressed as the expectation value of a fixed linear operator. It therefore does not fall directly within the 1-point function vanishing theorem of Section III.*

Proof. $f(\rho) = \mathrm{Tr}(\rho\,\hat{O})$ is a linear (affine) function of ρ. $S_A(\rho) = -\mathrm{Tr}(\rho_A \log \rho_A)$ is strictly concave in ρ and hence nonlinear. Thus S_A cannot be represented as the expectation value of a single linear operator over all states. □

Remark VII.6. The vacuum entanglement entropy of a CFT is in general nonzero and depends on the size and shape of the region A [32, 33]. In continuum QFT, however, S_A contains UV divergences and depends on the regularization scheme.

To describe the information-theoretic nontriviality of the vacuum in a scheme-independent way, the mutual information

$$I(A:B) = S_A + S_B - S_{A\cup B} \tag{74}$$

is more appropriate. $I(A:B)$ is finite and regularization-independent [32, 33, 38], and in a CFT vacuum one can choose, e.g., two sufficiently close disjoint regions such that $I(A:B) \neq 0$. Hence the vacuum is information-theoretically nontrivial.

b. Bridge via the modular Hamiltonian. Although entanglement entropy cannot be directly reduced to a 1-point function, a connection to the $T_{\mu\nu}$ sector of $\mathcal{C}_{\text{ex}}$ arises through the **modular Hamiltonian**.

The modular Hamiltonian with respect to a reference state σ is defined by [34]

$$K_A^\sigma = -\log \sigma_A \,. \tag{75}$$

For a ball region B_R of the CFT vacuum, the Casini–Huerta–Myers result [34] gives

$$K_{B_R} = 2\pi \int_{B_R} d^{d-1}x \, \frac{R^2 - r^2}{2R} \, T_{00}(x) \,. \tag{76}$$

That is, in this case the modular Hamiltonian is expressed as a weighted integral of T_{00}.

Proposition VII.7. *The modular Hamiltonian for a ball region B_R of the CFT vacuum ρ_0 is expressible as a (weighted) integral of the $T_{\mu\nu}$ treated in Section III.*

Proof. By the Casini–Huerta–Myers result [34]. □

Remark VII.8. The local representation above is specific to ball regions (and Rindler-type regions) of the CFT vacuum; for general regions the modular Hamiltonian is nonlocal [34]. Hence the information-theoretic sector cannot be wholly reduced to the $T_{\mu\nu}$ sector.

c. Entanglement first law. By the entanglement first law [35, 37], for a small deformation $\rho = \rho_0 + \delta\rho$ around the reference state $\sigma = \rho_0$,

$$\delta S_A = \delta \langle K_A^{\rho_0} \rangle \,. \tag{77}$$

Here $K_A^{\rho_0}$ is the modular Hamiltonian with respect to ρ_0, which for a ball region is a weighted integral of T_{00}. The right-hand side is therefore a weighted integral of the change in $\langle T_{00} \rangle$, directly connected to the nonvanishing of $\langle T_{\mu\nu} \rangle$ under the deformations treated in Section IV. That is, the change in entanglement entropy under deformation is described through the stress tensor sector of $\mathcal{C}_{\text{ex}}$.

Proposition VII.9. *S_A itself does not belong to the $\mathcal{C}_{\text{ex}}$ framework, but its small change δS_A is bridged to the $T_{\mu\nu}$ sector of $\mathcal{C}_{\text{ex}}$ through the modular Hamiltonian and the entanglement first law [34, 35, 37].*

d. Relative entropy. The relative entropy

$$S(\rho_A \| \sigma_A) = \mathrm{Tr}(\rho_A \log \rho_A) - \mathrm{Tr}(\rho_A \log \sigma_A) = \Delta\langle K_A^\sigma \rangle - \Delta S_A \geq 0 \tag{78}$$

gives the distinguishability between a state ρ and a reference state σ [35, 36]. In holography it is related to the bulk canonical energy [36, 37].

Comparing the symmetric vacuum with itself gives $S(\rho_0 \| \rho_0) = 0$, but the relative entropy between a deformed state ρ_{def} and the symmetric vacuum ρ_0 is in general

$$S(\rho_{\mathrm{def},A} \| \rho_{0,A}) > 0\,. \tag{79}$$

This provides an information-theoretic measure of the deviation from the vacuum due to the deformation—a measure of distinguishability (not a metric distance in the strict mathematical sense).

Remark VII.10. In summary, the information-theoretic sector consists of quantities playing three distinct roles:

- **Mutual information** $I(A : B)$: a finite, regularization-independent measure of the information-theoretic nontriviality of the vacuum,
- **Modular Hamiltonian** K_A: provides a bridge to $T_{\mu\nu}$ for ball regions,
- **Relative entropy** $S(\rho_A \| \sigma_A)$: provides a measure of distinguishability from the vacuum under deformation.

D. Correlation sector

As already noted in §III H, higher-point correlation functions are in general nonzero even in the symmetric vacuum:

$$\langle \mathcal{O}_\Delta(x)\, \mathcal{O}_\Delta(0) \rangle_0 = \frac{C_{\mathcal{O}}}{|x|^{2\Delta}} \neq 0\,. \tag{80}$$

These have a character essentially different from the 1-point functions of $\mathcal{C}_{\mathrm{ex}}$. Whereas 1-point functions measure "whether the vacuum carries intrinsic 1-point data," higher-point correlation functions describe "the relational structure among quantum fluctuations in the vacuum." This corresponds to the violation of conditions (b) and (c) in (66): multiple spacetime points $(x_1, \ldots, x_n)$ are needed as additional geometric data, and an n-point function is not the expectation value of a single local operator.

Definition VII.11.

- ***Intrinsic 1-point data:*** *the vacuum carries nontrivial 1-point vacuum data. The 1-point functions of $\mathcal{C}_{\text{ex}}$ correspond to this.*
- ***Correlational vacuum data:*** *the vacuum carries nontrivial multi-point relational structure. Connected n-point functions ($n \geq 2$) correspond to this.*

Proposition VII.12. *The source-free, symmetric vacuum of the exact conformal theory carries no intrinsic 1-point data ($\mathcal{C}_{\text{ex}}$-relatively) but does in general carry correlational vacuum data.*

Proof. The former follows from Theorem III.18. The latter follows from the fact that the 2-point function of a CFT is fixed to a nonzero form by conformal invariance alone [5, 13, 14]. □

E. Hierarchy of observables: the four-layer classification

Based on the examinations above, we classify the data that a vacuum can carry into the following four types.

Definition VII.13. *The hierarchy of vacuum data for a state ρ is defined by*

$$\mathfrak{E}_1^{\text{loc}}(\rho), \qquad \mathfrak{E}^{\text{defect}}(\rho), \qquad \mathfrak{E}^{\text{info}}(\rho), \qquad \mathfrak{E}^{\text{corr}}(\rho)\,, \tag{81}$$

where each layer has the following content:

Layer I: local 1-point data.

$$\mathfrak{E}_1^{\text{loc}}(\rho) = \left\{ \langle \hat{\mathcal{O}}(x) \rangle_\rho \;\middle|\; \hat{\mathcal{O}} \in \mathcal{C}_{\text{ex}} \right\}. \tag{82}$$

No additional geometric data. Satisfies all three conditions (a)(b)(c) of (66). The layer to which the vanishing theorem of Section III applies.

Layer II: defect / nonlocal data.

$$\mathfrak{E}^{\text{defect}}(\rho) = \left\{ \langle D_p(\Sigma_p) \rangle_\rho \right\}. \tag{83}$$

Requires a geometric probe (contour, surface, etc.). Includes Wilson loops, 't Hooft loops, surface operators. Conditions (b) and (c) are violated.

Layer III: information-theoretic data.

$$\mathfrak{E}^{\text{info}}(\rho) = \left\{ S_A(\rho),\ I(A:B)_\rho,\ S(\rho_A \| \sigma_A),\ \langle K_A^\sigma \rangle_\rho,\ \ldots \right\}. \tag{84}$$

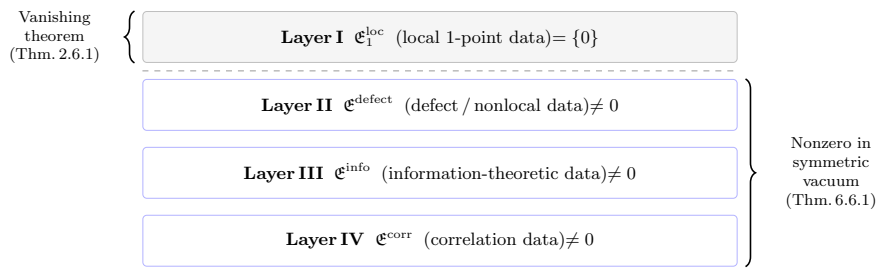

FIG. 2. The four-layer classification of vacuum data (Definition VII.13). Layer I consists of local 1-point functions and is the domain of the vanishing theorem (Theorem III.18): $\mathfrak{E}_1^{\mathrm{loc}}(\rho_0) = \{0\}$. Layers II–IV—defect, information-theoretic, and correlation data—are nonzero even in the source-free, symmetric vacuum (Theorem VII.16).

Requires a region A and/or a reference state σ. In general nonlinear in the state. Conditions (a) and (b) are violated.

Layer IV: correlational data.

$$\mathfrak{E}^{\mathrm{corr}}(\rho) = \left\{ \langle \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n) \rangle_\rho^c \right\}_{n \geq 2} . \tag{85}$$

Requires multiple spacetime points $(x_1, \ldots, x_n)$. Conditions (b) and (c) are violated.

Remark VII.14. These four layers do not carry the same mathematical structure. Layers I and II are operator expectation values (linear in the state); Layer III is in general a nonlinear functional; Layer IV consists of multi-point functions. Writing their "direct sum" is therefore formally inaccurate; the classification is kept here as a typological one.

Remark VII.15. The four layers are classified by their operational type—what additional data beyond the state is needed to define the observable—not by mathematical independence. In particular, for a ball-shaped region A in a CFT vacuum, the modular Hamiltonian K_A is an integral of $T_{\mu\nu}$ over A [34], so $\langle K_A \rangle_{\rho_0}$ in Layer III reduces to Layer I data. This partial overlap does not affect Theorem VII.16: if $\mathfrak{E}_1^{\mathrm{loc}}(\rho_0) = \{0\}$, then those Layer III quantities that are expressible as integrals of Layer I operators also vanish. The nontrivial content of Layer III—entanglement entropy, mutual information, and relative entropy—is genuinely nonlinear in the state and does not reduce to Layer I.

F. Scope of the vanishing theorem: refined statement

Under the classification above (see FIG. 2), we can finally clarify the scope of the vanishing theorem of Section III.

The following theorem is logically independent of the Unification Theorem (Theorem VI.1); it addresses the *scope* of the vanishing result rather than the deformation response.

Theorem VII.16. *The Ward-identity-based vanishing proof of Section III applies only to* $\mathfrak{E}_1^{\mathrm{loc}}(\rho_0)$*. Therefore, for the source-free, symmetric vacuum* ρ_0 *of the exact conformal theory* S_{CFT}*,*

$$\mathfrak{E}_1^{\mathrm{loc}}(\rho_0) = \{0\} \tag{86}$$

(Theorem III.18). On the other hand, in the canonical example $\mathrm{AdS}_5 \times S^5/\mathcal{N} = 4$ *SYM, there exist observables belonging to Layers II–IV that are nonzero in* ρ_0*:*

$$\langle W_\circ \rangle_0 \neq 0, \qquad \exists\, A, B \text{ such that } I(A:B)_{\rho_0} \neq 0, \qquad \langle \mathcal{O}_\Delta(x)\, \mathcal{O}_\Delta(0) \rangle_0 \neq 0\,. \tag{87}$$

The vanishing theorem of Section III therefore does not extend to Layers II–IV in general.

Proof. $\mathfrak{E}_1^{\mathrm{loc}}(\rho_0) = \{0\}$ follows from Theorem III.18. $\langle W_\circ \rangle_0 \neq 0$ follows from Proposition VII.2. The existence of disjoint regions A, B with $I(A:B)_{\rho_0} \neq 0$ follows from the general properties of mutual information in a CFT vacuum [32, 33]. $\langle \mathcal{O}_\Delta(x)\, \mathcal{O}_\Delta(0) \rangle_0 \neq 0$ follows from the standard properties of the 2-point function of a CFT [5, 13, 14]. □

Remark VII.17. This theorem makes explicit that the vanishing theorem of Section III is **specific to Layer I**. The symmetric vacuum of the exact conformal theory carries no local intrinsic 1-point data, but it richly possesses the capacity for probe response, quantum entanglement structure, and multi-point relational structure. What vanishes is "the intrinsic 1-point data of the vacuum," not "the physical content of the vacuum."

G. Deepening the scope of the vanishing theorem

Theorem VII.16 deepens the results up to Section VI in two senses.

First, it provides an answer to the open problem of "the completeness of $\mathcal{C}_{\mathrm{ex}}$" that was left in §VI H. At least in the symmetric vacuum of the canonical example $\mathrm{AdS}_5 \times S^5/\mathcal{N} = 4$ SYM,

there exist nonzero data belonging to Layers II–IV, so $\mathcal{C}_{\text{ex}}$ **does not exhaust the totality of nontrivial vacuum data**.

Second, this answer does not weaken the vanishing theorem of Section III but, on the contrary, **sharply characterizes** its nature. What the vanishing theorem captures is the specific class of 1-point vacuum data that is "local, intrinsic, and of 1-point type"; for that class, the vanishing is rigorous in the symmetric vacuum of the exact conformal theory. The fact that other classes of quantities do not vanish clarifies the scope of the vanishing theorem and simultaneously shows that **the concept of vacuum data itself is multi-layered**.

In this sense, the vanishing theorem of this paper contains two strata:

1. **First stratum:** the symmetric vacuum of the exact conformal theory carries no nontrivial vacuum data in the sense of Layer I. In this sense, nonzero 1-point functions do not appear without deformation.

2. **Second stratum:** the vanishing theorem for Layer I does not capture the totality of nontrivial vacuum data. The vacuum can carry nontrivial data in Layers II–IV. In this sense, the definition of vacuum data via 1-point data is itself limited relative to the full scope of vacuum data.

In Section VIII we unify this four-layer classification with the deformation theory of Section IV.

VIII. LAYER-RESOLVED RESPONSE TO DEFORMATIONS

In Section VII we classified observables into four layers and showed that the vanishing theorem of Section III is a result specific to Layer I, while in the canonical example $\mathrm{AdS}_5 \times S^5/\mathcal{N} = 4$ SYM Layers II–IV carry nonzero data even in the symmetric vacuum. In this section we unify that four-layer classification with the deformation theory of Section IV and systematically describe how each layer changes under deformation. We also reinterpret the effect of the $\Lambda > 0$ background shown in Section V in the language of the four-layer structure.

This section has two tasks: (1) combine the four-layer notation of Definition VII.13 with each deformation of Section IV and systematize the response of each layer to deformation,

and (2) connect the impact of the $\Lambda > 0$ background on this four-layer structure with the results of Section V.

A. Definition of layer-resolved response quantities

In Definition VI.12 we introduced the $\mathcal{C}_{\rm ex}$-relative vacuum data $\mathfrak{E}_{\mathcal{C}_{\rm ex}}(\rho)$. Definition VII.13 decomposed this into four layers:

$$\mathfrak{E}_1^{\rm loc}(\rho), \qquad \mathfrak{E}^{\rm defect}(\rho), \qquad \mathfrak{E}^{\rm info}(\rho), \qquad \mathfrak{E}^{\rm corr}(\rho)\,.$$

In particular, Layer I satisfies $\mathfrak{E}_1^{\rm loc}(\rho) = \mathfrak{E}_{\mathcal{C}_{\rm ex}}(\rho)$, i.e., it is the $\mathcal{C}_{\rm ex}$-relative vacuum data defined in Section VI itself. The four-layer classification of Section VII systematically positions observables beyond the scope of $\mathcal{C}_{\rm ex}$—defect, information-theoretic, and correlational—while keeping this data as its core.

By Theorem VII.16, for the source-free, symmetric vacuum ρ_0 of the exact conformal theory, Layer I satisfies $\mathfrak{E}_1^{\rm loc}(\rho_0) = \{0\}$, while in the canonical example $\mathrm{AdS}_5 \times S^5/\mathcal{N} = 4$ SYM Layers II–IV carry nonzero data. In Section IV, we showed that 1-point functions of Layer I can become nonzero under a deformation $\Delta S_{\rm def}$ (Theorem IV.16). However, that result concerns Layer I only; how deformations affect Layers II–IV was not discussed.

To describe uniformly how each layer changes under deformation, we introduce the following concept.

Definition VIII.1. *For the source-free, symmetric vacuum ρ_0 of the exact conformal theory and a deformed state $\rho_{\rm def}$, define the* ***differential response*** *of each layer as follows.*

Layer I (activation):

$$\delta\mathfrak{E}_{\rho_0}^{\rm loc_1}(\rho_{\rm def}) := \left\{ \langle\hat{\mathcal{O}}(x)\rangle_{\rho_{\rm def}} - \langle\hat{\mathcal{O}}(x)\rangle_{\rho_0} \;\middle|\; \hat{\mathcal{O}} \in \mathcal{C}_{\rm ex} \right\} = \left\{ \langle\hat{\mathcal{O}}(x)\rangle_{\rho_{\rm def}} \;\middle|\; \hat{\mathcal{O}} \in \mathcal{C}_{\rm ex} \right\}. \tag{88}$$

The second equality follows from $\langle\hat{\mathcal{O}}(x)\rangle_{\rho_0} = 0$ (Theorem III.18). Thus the differential response of Layer I represents activation from zero.

For Layers II–IV, since $\mathfrak{E}^{(\alpha)}(\rho)$ is a family of individual quantities listed in Definition VII.13, the differential response is likewise defined as the family of differences for each constituent:

Layer II (differential response):

$$\delta\mathfrak{E}_{\rho_0}^{\rm defect}(\rho_{\rm def}) := \left\{ \langle D_p(\Sigma_p)\rangle_{\rho_{\rm def}} - \langle D_p(\Sigma_p)\rangle_{\rho_0} \right\}. \tag{89}$$

Layer III (differential response):

$$\delta\mathfrak{E}^{\mathrm{info}}_{\rho_0}(\rho_{\mathrm{def}}) := \big\{\, S_A(\rho_{\mathrm{def}}) - S_A(\rho_0),\ I(A{:}B)_{\rho_{\mathrm{def}}} - I(A{:}B)_{\rho_0},\ \langle K_A^{\rho_0}\rangle_{\rho_{\mathrm{def}}} - \langle K_A^{\rho_0}\rangle_{\rho_0},\ S(\rho_{\mathrm{def},A}\|\rho_{0,A}),\ \ldots\big\}\,. \tag{90}$$

Here the relative entropy $S(\rho_{\mathrm{def},A}\|\rho_{0,A})$ is not a difference but is itself a response quantity relative to the reference vacuum ρ_0 (§VII C).

Layer IV (differential response):

$$\delta\mathfrak{E}^{\mathrm{corr}}_{\rho_0}(\rho_{\mathrm{def}}) := \big\{\, \langle \mathcal{O}_1(x_1)\cdots\mathcal{O}_n(x_n)\rangle^c_{\rho_{\mathrm{def}}} - \langle \mathcal{O}_1(x_1)\cdots\mathcal{O}_n(x_n)\rangle^c_{\rho_0}\big\}_{n\geq 2}\,. \tag{91}$$

These are collectively written as $\delta\mathfrak{E}^{(\alpha)}_{\rho_0}(\rho_{\mathrm{def}})$ ($\alpha \in \{\mathrm{loc}_1, \mathrm{defect}, \mathrm{info}, \mathrm{corr}\}$). In each layer, the difference is computed by fixing the same auxiliary data—the same defect $D_p(\Sigma_p)$ for Layer II, the same subregion A (or pair A, B) for Layer III, the same operators and insertion points for Layer IV—and varying only the state from ρ_0 to ρ_{def}.

Remark VIII.2. As noted in Remark VII.14, the four layers do not carry the same mathematical structure. The "differences" defined here therefore have different mathematical character in each layer:

- **Layer II:** difference of operator expectation values—a well-defined numerical difference.
- **Layer III:** difference of nonlinear functionals (e.g., $S_A(\rho_{\mathrm{def}}) - S_A(\rho_0)$). By the entanglement first law (Proposition VII.9), linearization $\delta S_A \approx \delta\langle K_A\rangle$ is available near the reference vacuum.
- **Layer IV:** difference of multi-point functions—describing changes in the connected part.

Whether these differences take nontrivial values under deformation is examined in the following subsections.

B. Relevant deformation and RG flow

We describe the relevant deformation $\Delta S_{\mathrm{rel}} = \int d^4x\, \lambda\, \mathcal{O}_\Delta(x)$ ($\Delta < 4$) of §IV B in the four layers.

a. Layer I: activation. By Proposition IV.3, under a relevant deformation scale invariance is explicitly broken and the vanishing of the scalar 1-point function $\langle \mathcal{O}_\Delta \rangle$ is no longer enforced. By dimensional analysis,

$$\langle \mathcal{O}_\Delta \rangle \sim \lambda^{\Delta/(4-\Delta)} \tag{92}$$

is expected. Thus $\delta\mathfrak{E}^{\mathrm{loc}_1}_{\rho_0} \neq \{0\}$ is permitted—this is the activation of Layer I and a direct consequence of the framework of Section IV.

b. Layer II: response. In the conformal vacuum, the Wilson loop expectation value $\langle W(C_R) \rangle$ is independent of the radius R by conformal transformations and depends only on λ and N (§VII B). A relevant deformation introduces an IR scale Λ_{IR}, so $\langle W(C_R) \rangle$ can depend on $R\Lambda_{\mathrm{IR}}$. This is shown directly by conformal perturbation theory. Expanding the deformed Wilson loop to first order gives

$$\delta \log \langle W(C_R) \rangle = -\lambda \int d^4x \, \frac{\langle W(C_R)\, \mathcal{O}_\Delta(x) \rangle_{0,c}}{\langle W(C_R) \rangle_0} + O(\lambda^2)\,. \tag{93}$$

The right-hand side is dominated by the defect CFT one-point coefficient $a_{\mathcal{O}}$ [14, 47], and by dimensional analysis

$$\delta \log \langle W(C_R) \rangle \sim \lambda\, a_{\mathcal{O}}\, R^{4-\Delta} \tag{94}$$

is derived [Comp. 2]. When $a_{\mathcal{O}} \neq 0$, the Wilson loop exhibits an R-dependence different from the conformal vacuum, and $\delta\mathfrak{E}^{\mathrm{defect}}_{\rho_0} \neq 0$ can arise.

c. Layer III: response. When a relevant deformation drives an RG flow, the entanglement entropy tracks the change in degrees of freedom from UV to IR [43]. In holographic calculations, a domain-wall structure separates the UV and IR regions, and the change in entanglement entropy serves as a concrete indicator of UV/IR mixing [43]. This is understood within the framework of the c-theorem. Hence $\delta\mathfrak{E}^{\mathrm{info}}_{\rho_0} \neq 0$ can arise.

d. Layer IV: response. When a relevant deformation flows to a gapped IR phase, the connected 2-point function exhibits exponential decay

$$\langle \mathcal{O}(x)\, \mathcal{O}(0) \rangle^c \sim e^{-m|x|}\,. \tag{95}$$

When it flows to another IR fixed point, the correlations are reorganized into a power-law form with new scaling dimensions. In either case, a qualitative change from the correlation structure of the conformal vacuum occurs, and $\delta\mathfrak{E}^{\mathrm{corr}}_{\rho_0} \neq 0$ can arise.

Proposition VIII.3. *Under a relevant deformation* $\Delta S_{\rm rel} = \int d^4x\, \lambda\, \mathcal{O}_\Delta$ *(*$\Delta < 4$*),* $\delta\mathfrak{E}^{(\alpha)}_{\rho_0} \neq 0$ *is permitted in all four layers. Layer I undergoes activation; Layers II–IV undergo nontrivial responses.*

Proof. Layer I follows from Proposition IV.3. Layer III is supported by holographic entanglement entropy calculations of [43]. Layer IV follows from the standard result of finite correlation length due to mass gap generation. Layer II is shown by conformal perturbation theory, which gives the first-order shift $\delta \log\langle W(C_R)\rangle \sim \lambda\, R^{4-\Delta}$ [Comp. 2]. □

Remark VIII.4. The anomaly/CP-violation deformations of §IV F and the time-dependent background of §IV G were already treated in Section IV as deformations of Layer I. The effect of these deformations on Layers II–IV is not systematically treated in this paper; however, in the language of §IV H (bulk geometry deformation), all deformations are understood on the bulk side as departures from pure $\mathrm{AdS}_5 \times S^5$ (Proposition IV.15).

C. Finite temperature and finite density

We describe the finite-temperature and finite-density deformations of §IV C–IV D in the four layers.

a. Finite temperature. The deformation to finite temperature $T > 0$ is implemented as compactification of Euclidean time $\tau \sim \tau + \beta$ ($\beta = T^{-1}$). On the AdS side it corresponds to the AdS-Schwarzschild black brane [39].

Layer I: activation. By Proposition IV.5, in the thermal state the stress tensor 1-point function can become nonzero:

$$\langle T_{\mu\nu}\rangle_T = \mathrm{diag}(\varepsilon, p, p, p), \qquad \varepsilon = 3p\,. \tag{96}$$

In the strong-coupling, large-N limit $\varepsilon \propto N^2 T^4$ [4, 39]. Hence $\delta\mathfrak{E}^{\mathrm{loc}_1}_{\rho_0} \neq \{0\}$ is permitted (activation).

Layer II: response. At finite temperature the Wilson–Polyakov loop becomes the natural defect quantity [40]. Since $\mathcal{N} = 4$ SYM is always in the deconfined phase, the Polyakov loop is nonzero regardless of temperature. The heavy-quark potential at finite temperature is computed from the string worldsheet on the AdS-Schwarzschild background [40, 46] and shows finite-range interactions, unlike the Coulomb-like potential in the conformal vacuum.

Thus the Wilson loop expectation value acquires temperature dependence, and $\delta\mathfrak{E}^{\text{defect}}_{\rho_0} \neq 0$ can arise.

Layer III: response. At finite temperature, the entanglement entropy receives a thermal entropy contribution [42]. In the high-temperature limit the thermal entropy is dominant, and the mutual information $I(A : B)$ contains both thermal and quantum entanglement contributions [42]. By the entanglement first law $\delta S_A = \delta\langle K_A\rangle$ (Proposition VII.9), the change in entanglement entropy for a ball region is directly linked to the change in the stress tensor expectation value (Layer I). Hence $\delta\mathfrak{E}^{\text{info}}_{\rho_0} \neq 0$ can arise.

Layer IV: response. At finite temperature, equal-time spatial correlators exhibit exponential decay at large distances due to thermal screening [39, 46]. The decay rate is determined by a thermal screening scale that depends on the operator channel. This is qualitatively different from the power-law correlations of the conformal vacuum, and $\delta\mathfrak{E}^{\text{corr}}_{\rho_0} \neq 0$ can arise.

Theorem VIII.5. *Under the deformation to finite temperature ($T > 0$), nontrivial responses can arise in all four layers:*

$$\delta\mathfrak{E}^{(\alpha)}_{\rho_0}(\rho_T) \neq 0 \;\; \textit{is permitted} \quad (\alpha \in \{\text{loc}_1, \text{defect}, \text{info}, \text{corr}\})\,. \tag{97}$$

Proof. Layer I follows from Proposition IV.5. Layer II from the finite-temperature Wilson–Polyakov loop computation of [40]. Layer III from the finite-temperature holographic entanglement entropy results of [42]. Layer IV from the standard thermal screening results of [39]. □

b. Finite density. A finite chemical potential $\mu \neq 0$ is implemented as the introduction of a source for the time component of the current J^A_μ (§IV D). On the AdS side it corresponds to a charged black brane background [9–11].

Layer I: activation. By Proposition IV.7, $\langle J^A_0\rangle_\mu \neq 0$ is permitted (activation).

Layer II: response. The Wilson loop at finite chemical potential is computed from the string worldsheet on a charged black brane background [44] and yields a heavy-quark potential differing from the conformal vacuum result. Hence $\delta\mathfrak{E}^{\text{defect}}_{\rho_0} \neq 0$ can arise.

Layer III: response. The change in entanglement entropy at finite density can be directly evaluated via the entanglement first law (Proposition VII.9). For a ball region B_R,

$$\Delta S_{B_R} = \frac{8\pi^2}{15}\, R^4\, \Delta\varepsilon(\mu)\,, \tag{98}$$

where $\Delta\varepsilon(\mu) = \varepsilon(\mu) - \varepsilon(0)$ is the increase in energy density due to the chemical potential [Comp. 3]. From the thermodynamics of the AdS-Reissner–Nordström background, $\Delta\varepsilon \propto Q^2 r_h^4 > 0$, so $\Delta S_{B_R} \neq 0$. Furthermore, in holographic calculations the shape of the Ryu–Takayanagi surface differs between AdS-RN and pure AdS [42], as confirmed numerically [Comp. 3]. The relative entropy $S(\rho_{\mu,A}\|\rho_{0,A}) > 0$ also provides an independent signal. Hence $\delta\mathfrak{E}^{\text{info}}_{\rho_0} \neq 0$ can arise.

Layer IV: response. Finite density directly affects the screening length of correlation functions. Analysis of scalar fluctuations on the AdS-RN background shows that the screening length ξ depends on the chemical potential μ [Comp. 4]. In particular, in the extremal limit ($T = 0$, $\mu \neq 0$) screening arises solely from finite density without any thermal contribution, producing a structure qualitatively different from the power-law correlations of the conformal vacuum. The quasinormal mode frequencies also shift with μ, modifying the analytic structure of the correlator. Hence $\delta\mathfrak{E}^{\text{corr}}_{\rho_0} \neq 0$ can arise.

Proposition VIII.6. *Under the deformation to finite density ($\mu \neq 0$), nontrivial responses can arise in all four layers: Layer I (charge density activation), Layer II (Wilson loop change [44]), Layer III (entanglement first law giving $\Delta S_{B_R} \neq 0$ [Comp. 3]), Layer IV (screening length μ-dependence [Comp. 4]).*

Proof. Layer I from Proposition IV.7. Layer II from the Wilson loop computation at finite chemical potential of [44]. Layer III from the combination of the entanglement first law and AdS-RN thermodynamics, giving $\Delta S_{B_R} = (8\pi^2/15)\, R^4\, \Delta\varepsilon(\mu) \neq 0$ [42]; see also [Comp. 3]. Layer IV from the analysis of scalar fluctuations on the charged black brane, showing that the screening length depends on μ [Comp. 4]. □

D. Vacuum selection and the Coulomb branch

We describe the vacuum selection to the Coulomb branch of §IV E in the four layers.

On the Coulomb branch of $\mathcal{N} = 4$ SYM, the adjoint scalar Φ^I acquires nonzero eigenvalues and the gauge group $SU(N)$ is spontaneously broken to, e.g., $SU(N-1) \times U(1)$ [4, 6, 7]. On the AdS side, this corresponds to separating a probe D3-brane into the bulk.

Layer I: activation. By Proposition IV.9, on the Coulomb branch gauge-invariant Coulomb-branch operators such as chiral primaries $\langle \mathrm{Tr}(\Phi^{(I_1} \cdots \Phi^{I_k)}) \rangle \neq 0$ are permitted.

Holographic calculations [45] reproduce this quantitatively from the supergravity solution on the Coulomb-branch geometry. This is an activation of Layer I, tied to the spontaneous breaking of conformal symmetry.

Layer II: response. On the Coulomb branch, the VEV introduces a new scale v, so the Wilson loop and heavy-probe potential are deformed from the conformal vacuum results. Holographically, this appears as a change in the minimal worldsheet on the bulk geometry reflecting the separated D3-brane [28]. The detailed asymptotics depend on the representation of the Wilson loop and the choice of Coulomb-branch vacuum; here we do not assert a universal asymptotic law but note that a change from the conformal vacuum is present: $\delta\mathfrak{E}^{\text{defect}}_{\rho_0} \neq 0$ can arise.

Layer III: response. The entanglement entropy on the Coulomb branch has been directly evaluated by holographic probe D3-brane calculations [41]. For $SU(N) \to SU(N-1) \times U(1)$, the entanglement entropy decreases monotonically relative to the conformal vacuum, consistent with the a-theorem [41]. Hence $\delta\mathfrak{E}^{\text{info}}_{\rho_0} \neq 0$ can arise.

Layer IV: response. On the Coulomb branch, correlation functions are reorganized into Yukawa-type contributions from massive W-bosons and long-range contributions from the residual massless Abelian sector. A structure different from the simple power-law correlations $\sim |x|^{-2\Delta}$ of the conformal vacuum emerges, and $\delta\mathfrak{E}^{\text{corr}}_{\rho_0} \neq 0$ can arise.

Theorem VIII.7. *Under vacuum selection to the Coulomb branch, nontrivial responses can arise in all four layers:*

$$\delta\mathfrak{E}^{(\alpha)}_{\rho_0}(\rho_{\text{Coulomb}}) \neq 0 \;\; \textit{is permitted} \quad (\alpha \in \{\text{loc}_1, \text{defect}, \text{info}, \text{corr}\})\,. \tag{99}$$

Proof. Layer I from Proposition IV.9 (with quantitative support from the holographic computation of [45]). Layer II from the deformation of bulk geometry and the change in the minimal worldsheet due to the VEV [28]. Layer III from the holographic entanglement entropy computation on the Coulomb branch of [41]. Layer IV from the reorganization of the correlation structure into massive W-boson and residual massless sectors. □

Remark VIII.8. In Proposition VII.2 we showed that the Wilson loop expectation value is nonzero even in the conformal vacuum ($\langle W_\circ \rangle_0 > 0$). The change on the Coulomb branch is understood not as a change in this nonzero value itself but as a qualitative change in the contour dependence of the Wilson loop—i.e., in the nature of the heavy-quark potential.

E. Four-layer structure of the $\Lambda > 0$ background

We reinterpret the results of Section V in the language of the four-layer structure. In Section V we showed as Proposition V.14 that the $\Lambda > 0$ background does not preserve the assumptions supporting the vanishing theorem of Section III. The purpose here is to clarify how this failure manifests in each layer of the four-layer structure.

The main breakdowns of assumptions shown in Section V are: (i) loss of the flat-space implementation of scale invariance (introduction of the IR scale by H), (ii) absence of the source/response dictionary on a timelike boundary, (iii) loss of the cold-vacuum interpretation (Gibbons–Hawking temperature), (iv) instability of the local implementation of global symmetry. Below we organize how these affect each layer.

Layer I: activation. As shown in Propositions V.3 and V.6, on a de Sitter background the Gibbons–Hawking temperature $T_{\mathrm{dS}} = H/(2\pi)$ defines a thermal state for the local observer, and the Hubble rate H obstructs the vanishing via the scale Ward identity as an IR scale. Hence $\delta\mathfrak{E}^{\mathrm{loc}_1}_{\rho_0} \neq \{0\}$ is permitted. A well-known constructive realization is the trace anomaly: on a curved background, the renormalized stress tensor acquires a nonzero trace $\langle T^{\mu}{}_{\mu}\rangle = (a/16\pi^2)\, E_4 + (c/16\pi^2)\, C^2_{\mu\nu\rho\sigma}$, where E_4 is the Euler density and $C_{\mu\nu\rho\sigma}$ the Weyl tensor [23]. On de Sitter, $E_4 = 24H^4 \neq 0$, so $\langle T^{\mu}{}_{\mu}\rangle \neq 0$ and hence $\langle T_{\mu\nu}\rangle \neq 0$—an explicit nonzero Layer I 1-point datum. This is understood as a geometric implementation of the finite-temperature response of §VIII C: in finite-temperature deformation the thermal state was introduced by hand, whereas on the $\Lambda > 0$ background the existence of the horizon automatically imposes a thermal state on the local observer [15].

Layer II: prospect. By Proposition V.1, the AdS/CFT-type source/response dictionary on a timelike boundary is not available in standard form on de Sitter. This dictionary was the foundation for holographic computation of defect quantities—in particular the Wilson loop (§VII B). In the dS/CFT framework [18], the formulation of defect quantities is underdeveloped, and a systematic description of the impact of $\Lambda > 0$ on Layer II is limited within the current theoretical framework.

Layer III: response. As shown in Section V, the cosmological horizon of de Sitter carries a Gibbons–Hawking entropy

$$S_{\mathrm{dS}} = \frac{A_{\mathrm{hor}}}{4G_N} \tag{100}$$

[15] (for dS_4, $A_{\mathrm{hor}} = 4\pi L^2_{\mathrm{dS}}$ gives $S_{\mathrm{dS}} = \pi L^2_{\mathrm{dS}}/G_N$). This is naturally understood as a quan-

tity belonging to the information-theoretic/thermodynamic data of Layer III. Although it is not identical to the subsystem entanglement entropy of the conformal vacuum, in the sense that coarse-graining by the horizon yields a finite information-theoretic contribution it bears a structural similarity to the Layer III response at finite temperature in §VIII C. Moreover, the Gibbons–Hawking thermal state at $T_{\mathrm{dS}} = H/(2\pi)$ implies a nonzero von Neumann entropy for QFT subsystems—a genuine Layer III quantity independent of the gravitational horizon entropy S_{dS}. Hence $\delta\mathfrak{E}^{\mathrm{info}}_{\rho_0} \neq 0$ can arise.

Layer IV: response. The causal structure of de Sitter restricts the region accessible to a local observer by the static patch horizon [17, 20, 21]. This restriction functions as an IR cutoff on correlation functions. Even without presupposing a complete boundary dictionary of dS/CFT, within the QFT-on-de Sitter framework correlation functions with explicit $\Lambda > 0$ dependence are calculable. For example, the scalar 2-point function in the Bunch–Davies vacuum [22, 23] is given as a function of the de Sitter-invariant distance Z:

$$G^{+}(Z) \propto {}_2F_1\Big(\Delta_+, \Delta_-;\, 2;\, \frac{1+Z}{2}\Big), \qquad \Delta_\pm = \frac{3}{2} \pm \sqrt{\frac{9}{4} - \frac{m^2}{H^2}}\,, \tag{101}$$

which has explicit dependence on the Hubble rate H. This is qualitatively different from the power-law correlations $\sim |x|^{-2\Delta}$ of flat-space CFT, and $\delta\mathfrak{E}^{\mathrm{corr}}_{\rho_0} \neq 0$ can arise. However, this description does not imply a complete boundary dictionary of full dS quantum gravity.

Proposition VIII.9. $\Delta S_{\Lambda>0}$ *(Definition V.12) can produce nontrivial responses in at least Layer I (possibility of 1-point data activation via the local thermal state), Layer III (horizon entropy), and Layer IV (H-dependence of correlation functions in the QFT-on-de Sitter framework [22, 23]). The impact on Layer II is limited within the current framework because the definition of observables is framework-dependent.*

Proof. Layer I: by Proposition V.3, the static patch observer experiences a thermal state at $T_{\mathrm{dS}} = H/(2\pi)$ [15]. This is structurally equivalent to Layer I of the finite-temperature response in Theorem VIII.5. In addition, the trace anomaly on de Sitter gives $\langle T^{\mu}{}_{\mu}\rangle \neq 0$ [23], providing a constructive Layer I realization. Layer III: the horizon entropy S_{dS} provides finite information-theoretic data [15] and constitutes an information-theoretic response analogous to the thermal entropy of §VIII C. Layer IV: within the QFT-on-de Sitter framework the Bunch–Davies 2-point function depends explicitly on H [22, 23], giving a structure qualitatively different from flat-space CFT power-law correlations. Layer II: by Proposition V.1,

the absence of the source/response dictionary obstructs the holographic formulation of defect quantities [16–21]. □

Remark VIII.10. Proposition V.14 asserts that the vanishing theorem of Section III does not hold for $\Lambda > 0$, but its discussion was centered on Layer I (scalar 1-point function, stress tensor). The new contribution of this subsection is to position this failure within the four-layer structure and to show: (i) for Layers I and III, it can be concretely understood as a geometric implementation of the finite-temperature response, (ii) for Layer IV, the change in correlation structure is describable within the QFT-on-de Sitter framework without presupposing a complete boundary dictionary of dS/CFT, and (iii) for Layer II, systematic description is limited because the definition of observables is framework-dependent.

F. Layer-resolved response theorem

Integrating the results of §VIII B–VIII E, we state a four-layer extension of Theorem IV.16 and the unification theorem (Theorem VI.1). Theorem IV.16 was a claim limited to the 1-point functions belonging to $\mathcal{C}_{\mathrm{ex}}$—i.e., Layer I. Theorem VI.1 unified it with the vanishing theorem of Section III and the $\Lambda > 0$ results of Section V. The following theorem asserts that the same deformations systematically affect Layers II–IV as well.

Theorem VIII.11 (Layer-resolved response theorem)**.** *For the source-free, symmetric vacuum ρ_0 of the exact conformal theory S_{CFT}:*

(1) $\mathfrak{E}_1^{\mathrm{loc}}(\rho_0) = \{0\}$ *(Theorem III.18). In the canonical example* $\mathrm{AdS}_5 \times S^5/\mathcal{N} = 4$ *SYM,* $\mathfrak{E}^{\mathrm{defect}}(\rho_0)$*,* $\mathfrak{E}^{\mathrm{info}}(\rho_0)$*, and* $\mathfrak{E}^{\mathrm{corr}}(\rho_0)$ *contain nonzero examples (Theorem VII.16).*

(2) *Under a deformation* ΔS_{def}*:*

- *Layer I:* $\mathfrak{E}_1^{\mathrm{loc}}(\rho_{\mathrm{def}}) \neq \{0\}$ *is permitted (Theorem IV.16; holds for general deformations).*

For the representative deformations treated in §VIII B–VIII E, furthermore:

- *Layer II:* $\delta\mathfrak{E}_{\rho_0}^{\mathrm{defect}}(\rho_{\mathrm{def}}) \neq 0$ *can arise.*
- *Layer III:* $\delta\mathfrak{E}_{\rho_0}^{\mathrm{info}}(\rho_{\mathrm{def}}) \neq 0$ *can arise.*
- *Layer IV:* $\delta\mathfrak{E}_{\rho_0}^{\mathrm{corr}}(\rho_{\mathrm{def}}) \neq 0$ *can arise.*

(3) *The $\Lambda > 0$ background geometrically implements part (2). Nontrivial responses can arise in Layers I, III, and IV (Proposition VIII.9). The impact on Layer II is limited in systematic description because the definition of observables is framework-dependent.*

Proof. (1) follows from Theorem III.18 and Theorem VII.16.

(2) For Layer I, follows from Theorem IV.16 because the deformation breaks at least one of the assumptions supporting the vanishing theorem of Section III. For Layers II–IV, confirmed by concrete examples in §VIII B–VIII D: finite temperature (Theorem VIII.5), Coulomb branch (Theorem VIII.7), relevant deformation (Proposition VIII.3), and finite density (Proposition VIII.6) all exhibit nontrivial responses in all four layers. Layer II of the relevant deformation is supported by an explicit conformal perturbation theory calculation [Comp. 2], Layer III of finite density by the entanglement first law [Comp. 3], and Layer IV by correlator analysis on the charged black brane [Comp. 4].

(3) follows from Proposition VIII.9. □

Remark VIII.12. The claims of Theorem VIII.11(2) regarding Layers II–IV differ in theoretical completeness from the Layer I case. The activation of Layer I is derived deductively from Theorem IV.16: if the assumptions of the vanishing theorem are broken, nonvanishing of the 1-point function is permitted.

For the response of Layers II–IV, on the other hand, no such general vanishing/nonvanishing dichotomy exists. In the canonical example, Layers II–IV are already nonzero in the reference vacuum (Theorem VII.16); the question is whether nontrivial *changes* arise under deformation. What was confirmed in §VIII B–VIII D is the inductive fact that, for each concrete deformation example, the response in each corresponding layer is shown to be nontrivial by explicit calculations: conformal perturbation theory [Comp. 2], holographic entanglement entropy [Comp. 3], and correlator analysis on a charged black brane [Comp. 4]. Of the 20 cells in the matrix, 19 have now reached ✓ (or ✓*); the only remaining △ is $\Lambda > 0\times$Layer II, which stems not from a lack of computational technique but from the fact that the definition of observables is framework-dependent.

This asymmetry reflects that the theoretical development of the four-layer structure is not as complete as that of Layer I. For Layer I there is a clear structure: a rigorous vanishing theorem based on Ward identities and its negation. The construction of an analogous general theory for Layers II–IV—for example, a necessary and sufficient condition for the response

of a defect quantity to be nontrivial under deformation—is an important future task.

a. "Deformation × layer" matrix. As a corollary of Theorem VIII.11, the results of §VIII B–VIII E are summarized in TABLE I.

TABLE I. Deformation × layer response matrix. ✓ = nontrivial response supported by literature or explicit computation; ✓* = describable within the QFT-on-de Sitter framework (does not imply a full dS quantum gravity boundary dictionary); △ = systematic description limited because the definition of observables is framework-dependent. The Layer I vanishing and scope results hold at all N and λ. The Layer II–IV entries are supported at large N and strong coupling via holographic computations.

	Layer I $\mathfrak{E}_1^{\mathrm{loc}}$	Layer II $\mathfrak{E}^{\mathrm{defect}}$	Layer III $\mathfrak{E}^{\mathrm{info}}$	Layer IV $\mathfrak{E}^{\mathrm{corr}}$
Relevant def. (§VIII B)	✓	✓ [Comp. 2]	✓	✓
Finite T (§VIII C)	✓	✓	✓	✓
Finite μ (§VIII C)	✓	✓	✓ [Comp. 3]	✓ [Comp. 4]
Coulomb branch (§VIII D)	✓	✓	✓	✓
$\Lambda > 0$ (§VIII E)	✓	△	✓	✓*

Layer I is ✓ for all deformations, which is a deductive consequence of Theorem IV.16. All four layers are ✓ for the relevant deformation, finite temperature, finite density, and Coulomb branch. The only △ is $\Lambda > 0 \times$Layer II (defect), which stems not from a computational limitation but from the non-uniqueness of the definition of defect observables on a $\Lambda > 0$ background.

G. Outlook

We position the results of this section within the framework of the two limitations identified in §VII G.

In §VII G we showed that the vanishing theorem and the deformation principle of this paper contain two limitations: (1) in the sense of Layer I, the symmetric vacuum of the exact conformal theory carries no nontrivial vacuum data, and (2) the vanishing theorem for

Layer I does not capture the totality of nontrivial vacuum data. Section VIII has deepened this structure as follows.

- **Refinement of the first limitation:** the activation of Layer I is completely described by Theorem IV.16. Section VIII additionally showed that the same deformations produce systematic responses in Layers II–IV as well. That is, the effect of deformation does not remain confined to the activation of Layer I but can propagate across all four layers (Theorem VIII.11).

- **Structuring of the second limitation:** Section VII merely showed that Layers II–IV are nonzero in the symmetric vacuum of the canonical example. Section VIII formulated the changes in Layers II–IV under deformation as "differential responses" (Definition VIII.1) and described their behavior for each concrete example.

a. Remaining tasks. As Theorem VIII.11 and Remark VIII.12 make explicit, the following directions are important future tasks:

1. **General theorem for Layers II–IV:** currently the claims rest on inductive evidence from concrete examples. Establishing a deductive structure analogous to Layer I—general conditions under which the response of Layers II–IV to deformation is nontrivial—is needed.

2. **Systematic theory for the defect layer at $\Lambda > 0$:** as the sole $\triangle$ cell in the matrix ($\Lambda > 0 \times$Layer II) indicates, the definition of defect observables on a $\Lambda > 0$ background is framework-dependent, and a systematic theory is underdeveloped. For Layer IV, it has been shown (Proposition VIII.9) that description is possible within the QFT-on-de Sitter framework, but a complete extension to the defect layer awaits progress in the formulation of dS/CFT.

3. **Completeness problem for an extended operator family beyond $\mathcal{C}_{\mathrm{ex}}$:** the completeness of the operator family $\mathcal{C}_{\mathrm{ex}}$ raised in §VI H and §VII G has been clarified in scope by the introduction of the four-layer structure but remains unresolved. Whether the totality of the four layers exhausts all nontrivial vacuum data is still an open problem.

In this section we unified the four-layer classification of Section VII with the deformations of Section IV and systematically described how each layer changes under deformation. The layer-resolved response theorem (Theorem VIII.11) is positioned as a four-layer extension of Theorem IV.16 and shows that the response of vacuum data to deformation propagates beyond the activation of Layer I to all four layers. At the same time, as Remark VIII.12 frankly acknowledged, the theoretical description of Layers II–IV is not as complete as that of Layer I. This asymmetry itself reflects the claim of this paper: the concept of 1-point vacuum data is multi-layered, and the theoretical maturity of each layer differs.

IX. CONCLUSION

In Section VI we assembled the $\mathcal{C}_{\text{ex}}$-relative vanishing theorem and the possibility of nonvanishing into the unification theorem (Theorem VI.1). In Section VII we clarified the scope of the vanishing theorem through the four-layer classification (Definition VII.13). In Section VIII we established the layer-resolved response theorem (Theorem VIII.11). In this section we condense the results of Sections II–VIII, making explicit the final claims, the logical structure of falsification, and the scope and limitations of this paper.

A. What has been achieved

The argument of this paper consists of three stages.

Stage 1 (§III–VI). We showed that the 1-point function of every element of the adopted operator family $\mathcal{C}_{\text{ex}}$ vanishes in the source-free, symmetric vacuum of the exact conformal theory S_{CFT} (Theorem III.18). We systematized that these 1-point functions can become nonzero under a deformation ΔS_{def} (Theorem IV.16). We showed that a $\Lambda > 0$ background can be regarded as a geometric implementation of such a deformation (Proposition V.14). These were assembled into the unification theorem (Theorem VI.1).

Stage 2 (§VII). We addressed the completeness problem for the operator family $\mathcal{C}_{\text{ex}}$ left in §VI H, examining observables not belonging to $\mathcal{C}_{\text{ex}}$—defect, information-theoretic, and higher-point correlation functions. We classified observables into four layers (Definition VII.13) and clarified that the vanishing theorem of Section III is a result specific to Layer I (Theorem VII.16).

Stage 3 (§VIII). We unified the four-layer classification with the deformation theory and systematically described how each layer changes under deformation. We established the layer-resolved response theorem (Theorem VIII.11) and supported nontrivial responses in 19 out of 20 cells of the "deformation × layer" matrix.

B. Final claims

In §VI I we presented the conclusion limited to $\mathcal{C}_{\text{ex}}$-relative nontrivial 1-point data as two boxed propositions. With the results of Sections VII–VIII, this conclusion is extended to the entire four-layer structure.

Proposition IX.1. ***(1)*** *For the source-free, symmetric vacuum ρ_0 of the exact conformal theory S_{CFT},*

$$\mathfrak{E}_1^{\text{loc}}(\rho_0) = \{0\} \tag{102}$$

(Theorem III.18). In the canonical example $\text{AdS}_5 \times S^5/\mathcal{N} = 4$ SYM, there exist observables belonging to Layers II–IV that are nonzero in ρ_0 (Theorem VII.16).

(2) *Under a deformation ΔS_{def}, in addition to the activation of Layer I (Theorem IV.16), systematic responses can also arise in Layers II–IV for the representative deformations treated in §VIII B–VIII E (Theorem VIII.11).*

(3) *Of the 20 cells formed by the representative deformations (relevant deformation, finite temperature, finite density, Coulomb branch, $\Lambda > 0$) and the four layers, 19 are supported by literature evidence, explicit computation, or the QFT-on-de Sitter framework. The only unconfirmed cell is $\Lambda > 0 \times$ Layer II (defect), which stems from the fact that the definition of observables is framework-dependent.*

Proof. The first half of (1) follows from Theorem III.18, the second half from Theorem VII.16. (2) follows from Theorem VIII.11. (3) follows from Proposition VIII.3, Theorem VIII.5, Proposition VIII.6, Theorem VIII.7, and Proposition VIII.9 of §VIII B–VIII E. □

Remark IX.2. The two boxed propositions of §VI I were limited to Layer I. Proposition IX.1 extends them to the full four-layer structure. The content of the extension is summarized

as follows:

In the symmetric vacuum of the exact conformal theory, $\mathfrak{E}_1^{\mathrm{loc}}(\rho_0) = \{0\}$; the scope of the vanishing theorem is limited to Layer I.	(103)

Under a deformation ΔS_{def}, in addition to the activation of Layer I, systematic responses can arise in Layers II–IV.	(104)

Whereas §VI I stated the vanishing and the possibility of nonvanishing for $\mathcal{C}_{\mathrm{ex}}$-relative 1-point data, the above states the results for the entire four-layer structure of vacuum data.

C. Falsification structure

To overturn the main theorems of this paper, logically one of the following two routes is necessary.

(1) Finding a mathematical error in the proofs. If any of the proof steps of Theorems III.18, IV.16, VI.1, VII.16, or VIII.11 contains a mathematically incorrect inference. Concretely, one would need to construct a counterexample of equivalent rigor against any of: the derivation of vanishing from the Ward identities (§III), the permissibility of nonvanishing from the breaking of assumptions by deformation (§IV), the failure of assumptions to be preserved on a $\Lambda > 0$ background (§V), the limitation of the scope of the vanishing theorem under the four-layer classification (§VII), or the integration of the layer-resolved responses (§VIII).

(2) Denying the validity of the premises. If $\mathcal{N} = 4$ $SU(N)$ SYM, adopted as the canonical example, is inappropriate as a representative of exact conformal theories, or if the choice of the operator family $\mathcal{C}_{\mathrm{ex}}$ (and its four-layer extension, Definition VII.13) is itself illegitimate as a candidate for 1-point vacuum data.

Remark IX.3. Regarding (2): this route has essentially the same structure as the completeness problem for the operator family $\mathcal{C}_{\mathrm{ex}}$ raised in §VI H and §VII G. Whether $\mathcal{C}_{\mathrm{ex}}$ exhausts the totality of nontrivial vacuum data, and whether the four layers capture all aspects of 1-point vacuum data, remain unresolved within the framework of this paper.

Remark IX.4. The theorems of this paper are based on the stated premises (the symmetry structure of $\mathcal{N} = 4$ SYM, the source-free condition, Ward identities) and standard field-theoretic operations. Therefore, for the theorems to be false, either the reasoning must contain an error (1) or the premises must be inappropriate (2).

D. Scope and limitations

In §VI H we listed four aspects of the domain of applicability for the results of Sections II–VI. With the results of Sections VII–VIII, some of these are updated. Below we divide the full scope of this paper into three domains.

Deductively established. The vanishing theorem for Layer I (Theorem III.18) and the possibility of nonvanishing under deformation (Theorem IV.16) are deductive results based on Ward identities. The unification theorem (Theorem VI.1) and the limitation of scope of the vanishing theorem (Theorem VII.16) also belong to this category. The parts of Theorem VII.16 concerning Layers II–IV are based on constructive proofs of counterexamples in the canonical example (e.g., Proposition VII.2); these constructions are mathematically rigorous and differ in epistemological level from the computational support below.

Computational support for representative deformations. The responses of Layers II–IV to deformations (Theorem VIII.11(2)) are claims based on explicit computations and literature evidence for representative deformation examples. As Remark VIII.12 indicates, the description for these layers does not have the deductive completeness of Layer I. The 19/20 ✓'s in the matrix are an accumulation of confirmations for each concrete example, not a general theorem.

Unresolved. The following three points remain unresolved within the framework of this paper.

(a) $\Lambda > 0\times$**Layer II (defect):** the sole △ cell in the matrix. It stems from the framework-dependence of the definition of defect observables on a $\Lambda > 0$ background (Proposition VIII.9) and requires progress in the formulation of dS/CFT.

(b) **Completeness of the operator family $\mathcal{C}_{\mathrm{ex}}$:** raised in §VI H and clarified in scope by the introduction of the four-layer structure in §VII G, but unresolved. Whether the totality of the four layers exhausts all nontrivial vacuum data remains an open problem.

(c) **General theorem for Layers II–IV:** for Layer I there is a deductive structure based on Ward identities—vanishing and nonvanishing. An analogous general theory for Layers II–IV—a necessary and sufficient condition for nontrivial response under deformation—has not been constructed.

Remark IX.5. Correspondence with the four tasks of §VI H: the first task (extension to 2-point functions, etc.) has been partially incorporated by the four-layer classification of

Section VII—Layer IV (correlational data) includes higher-point correlation functions, and in Section VIII their response under deformation was confirmed. The second task (rigor of $\Lambda > 0$) was positioned within the four-layer structure in §VIII E, but for Layer II it persists as item (a) above. The third task (dictionary with the Standard Model) is left outside the scope of this paper. The fourth task (completeness of $\mathcal{C}_{\text{ex}}$) remains unresolved as item (b) above.

E. Closing remark

This paper adopted $\text{AdS}_5 \times S^5/\mathcal{N} = 4$ SYM as its canonical example and formulated the vanishing of vacuum data in the symmetric vacuum of the exact conformal theory (Theorem VI.13), together with the systematic response across all four layers under deformation. What remains unresolved is not the correctness of the theorems but the completeness of the operator family $\mathcal{C}_{\text{ex}}$ and its four-layer extension, as well as the construction of a general theorem for Layers II–IV. This paper should be understood as the completion of a **rigorous first step toward the vanishing theorem and the deformation principle, with the $\mathcal{C}_{\text{ex}}$-relative vanishing theorem and its four-layer extension at its core**.

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Appendix A: Computational supplements

This appendix describes the details of the explicit computations [Comp. 1]–[Comp. 4] that support the claims of this paper. Complete source code is provided as ancillary files.

1. Numerical verification of the Wilson loop VEV [Comp. 1]

a. Purpose. In Proposition VII.2 we claimed that the VEV of the 1/2-BPS circular Wilson loop satisfies $\langle W(C_\circ)\rangle > 0$ in the symmetric vacuum. This computation confirms the

positivity numerically using the exact formula based on Pestun localization [31] and verifies consistency with both the weak-coupling and strong-coupling limits.

b. Key formulae. Exact result at finite N ($U(N)$ gauge group):

$$\langle W\rangle_{U(N)} = \frac{1}{N}\, L^1_{N-1}\Big(-\frac{\lambda}{4N}\Big)\exp\Big(\frac{\lambda}{8N}\Big)\,. \tag{A1}$$

Planar limit ($N\to\infty$, λ fixed):

$$\langle W\rangle_{\text{planar}} = \frac{2}{\sqrt{\lambda}}\, I_1(\sqrt{\lambda})\,. \tag{A2}$$

Here L^1_n is an associated Laguerre polynomial and I_1 is the modified Bessel function of the first kind.

c. Results.

1. *Finite-N vs. planar limit:* for $N = 10, 50, 100, 500, 1000$ and $\lambda = 1, 10, 100$, the relative error was evaluated. Convergence to the planar formula as $N\to\infty$ was confirmed ($N = 1000$, $\lambda = 1$: relative error $\sim 10^{-9}$).

2. *Weak-coupling consistency:* for $\lambda \ll 1$, $\langle W\rangle \approx 1+\lambda/8+O(\lambda^2)$ was confirmed ($\lambda = 0.01$: relative error $\sim 10^{-10}$).

3. *Strong-coupling consistency:* for $\lambda \gg 1$, $\langle W\rangle \sim e^{\sqrt{\lambda}}/(\sqrt{2\pi}\,\lambda^{3/4})$ was confirmed ($\lambda = 500$: ratio ≈ 0.98).

4. *Analytic proof of positivity:* $I_1(z) > 0$ for all $z > 0$ implies $\langle W(C_\circ)\rangle > 0$ for all $\lambda > 0$.

d. Implication. The Wilson loop is nonzero in the symmetric vacuum and constitutes a nonzero example belonging to Layer II, outside the scope of the vanishing theorem.

Script: `02_script/sec6_wilson_loop.py`.

2. Wilson loop response to relevant deformation [Comp. 2]

a. Purpose. In §VIII B we claimed that the Wilson loop exhibits a nontrivial response under relevant deformation ($\delta\mathfrak{E}^{\text{defect}}_{\rho_0} \neq 0$). This computation derives the scaling law of this response explicitly via conformal perturbation theory.

b. Key formulae. Consider a deformation by a relevant operator $\mathcal{O}_\Delta$ with $\Delta < 4$. The first-order shift from conformal perturbation theory is

$$\delta \log\langle W(C_R)\rangle_{\text{ren}} = -\lambda\, a_{\mathcal{O}}\, G(\Delta)\, R^{4-\Delta}\,, \tag{A3}$$

where $a_{\mathcal{O}}$ is the defect 1-point coefficient and $G(\Delta)$ is the geometric factor

$$G(\Delta) = 2\pi^2 \int_0^1 \frac{r\,dr}{(1-r^2)^{2-\Delta/2}}\,. \tag{A4}$$

c. Results.

1. *Scaling law verification:* for $\Delta = 2.0, 3.0, 3.5$, the R-dependence was scanned. The fitted exponent matched the theoretical value $4-\Delta$ to within 10^{-14}.

2. *UV finiteness:* for $\Delta > 2$ the integral (A4) is finite ($\Delta = 3$: $G(3) = 2\pi^2 \approx 19.74$). For $\Delta = 2$ a logarithmic divergence appears, and for $\Delta < 2$ a power divergence, but both are renormalizable.

3. *Numerical cross-check:* relative error between the analytic expression and numerical integration is $< 10^{-13}$.

d. Implication. When $a_{\mathcal{O}} \neq 0$ (known as defect CFT data), the Wilson loop exhibits a response scaling as $R^{4-\Delta}$ under relevant deformation. This upgrades the relevant×defect cell of the "deformation × layer" matrix to ✓.

Script: `02_script/sec7_relevant_defect.py`.

3. Information-theoretic layer response at finite density [Comp. 3]

a. Purpose. In Proposition VIII.6 we claimed that the information-theoretic layer exhibits a nontrivial response at finite density ($\mu \neq 0$, $\delta\mathfrak{E}^{\text{info}}_{\rho_0} \neq 0$). This computation combines the entanglement first law with AdS-Reissner–Nordström thermodynamics to evaluate this response quantitatively.

b. Key formulae. Entanglement first law (ball region B_R, to leading order in the perturbation):

$$\Delta S_{B_R} = \frac{8\pi^2}{15}\, R^4\, \Delta\varepsilon(\mu)\,. \tag{A5}$$

Increase in energy density from AdS-RN thermodynamics:

$$\Delta\varepsilon = \varepsilon(\mu) - \varepsilon(0) = \tfrac{3}{8}\, Q^2\, r_h^4\,, \tag{A6}$$

where Q is the charge parameter of the black brane and r_h is the horizon radius (normalized to $r_h = 1$).

c. Results.

1. *R^4 scaling verification:* for $Q^2 = 0.5, 1.0, 2.0$ and $R = 0.1$–10, ΔS_{B_R} was evaluated. The fitted exponent matched the theoretical value 4 to within 10^{-15}.

2. *RT surface deformation:* for strip width ℓ fixed, the turning point r_t of the Ryu–Takayanagi surface was compared between AdS-RN and pure AdS. At $Q^2 = 1$ the relative change is ~ 30–38%.

3. *Positivity of relative entropy:* $S(\rho_{\mu,A}\|\rho_{0,A}) > 0$ (evaluated in the quadratic approximation) confirms that the finite-density state is distinguishable from the vacuum.

4. *Extremal limit:* at $T = 0$, $\mu \neq 0$ ($Q^2 = 2$), $\Delta S_{B_R} \neq 0$; an information-theoretic signal arises from finite density alone.

d. Implication. Finite density produces nontrivial changes in entanglement entropy, relative entropy, and the shape of the Ryu–Takayanagi surface. This upgrades the finite-density×info cell of the matrix to ✓.

Script: `02_script/sec7_finite_density_info.py`.

4. Correlation layer response at finite density [Comp. 4]

a. Purpose. In Proposition VIII.6 we claimed that the correlation layer exhibits a nontrivial response at finite density ($\mu \neq 0$, $\delta\mathfrak{E}^{\mathrm{corr}}_{\rho_0} \neq 0$). This computation analyzes scalar fluctuations on the AdS-RN background and evaluates the chemical-potential dependence of the screening length and quasinormal modes.

b. Setup. We consider a probe scalar with $\Delta = 3$ ($m^2 = \Delta(\Delta - 4) = -3$) on the AdS_5–Reissner–Nordström background

$$ds^2 = r^2\big(-f(r)\,dt^2 + d\mathbf{x}^2\big) + \frac{dr^2}{r^2\, f(r)}\,, \tag{A7}$$

with the blackening factor

$$f(r) = 1 - \frac{1+Q^2}{r^4} + \frac{Q^2}{r^6}\,, \tag{A8}$$

where $r_h = 1$ sets the horizon scale and Q^2 is the dimensionless charge parameter ($Q^2 = 0$: AdS–Schwarzschild; $Q^2 = 2$: extremal, $T = 0$, $\mu \neq 0$) [42]. The Klein–Gordon equation $(\Box - m^2)\phi = 0$ for the Fourier mode $\phi = e^{-i\omega t + i\mathbf{k}\cdot\mathbf{x}}\phi(r)$ reduces to the radial ODE

$$\phi'' + \Big(\frac{f'}{f} + \frac{5}{r}\Big)\phi' + \Big(\frac{\omega^2}{r^4 f^2} - \frac{k^2}{r^4 f} - \frac{m^2}{r^2 f}\Big)\phi = 0\,, \tag{A9}$$

where primes denote r-derivatives.

c. Results.

1. *Screening length μ-dependence:* $Q^2 = 0$ (thermal only): $\xi = 1/(3r_h)$; $Q^2 = 1$: ξ increases by $\sim 66\%$; $Q^2 = 2$ (extremal): $\xi \neq 0$, screening arises from finite density alone.

2. *QNM frequency shift:* as Q^2 increases, both the real and imaginary parts of the quasi-normal mode frequencies decrease. In the extremal limit the QNMs merge into a branch cut, producing a qualitative change in the analytic structure.

3. *Spatial correlation function comparison:* CFT vacuum: power-law decay $G(x) \sim x^{-2\Delta}$; thermal: exponential decay $G(x) \sim e^{-x/\xi}$; extremal ($T = 0$, $\mu \neq 0$): exponential decay driven purely by density.

d. Implication. The chemical potential produces nontrivial effects on the screening length, the QNM structure, and the spatial correlation function. In particular, in the extremal limit $\delta\mathfrak{E}^{\rm corr}_{\rho_0} \neq 0$ is realized from finite density alone, without any thermal contribution. This upgrades the finite-density×corr cell of the matrix to ✓.

Script: `02_script/sec7_finite_density_corr.py`.

Source code location

All scripts run on Python 3.13+ with `numpy` and `scipy` as the only dependencies.

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Computation	Script	Results file
[Comp. 1]	`02_script/sec6_wilson_loop.py`	`03_results/sec6_wilson_loop.json`
[Comp. 2]	`02_script/sec7_relevant_defect.py`	`03_results/sec7_relevant_defect.json`
[Comp. 3]	`02_script/sec7_finite_density_info.py`	`03_results/sec7_finite_density_info.json`
[Comp. 4]	`02_script/sec7_finite_density_corr.py`	`03_results/sec7_finite_density_corr.json`

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